Minority Stakes, Technology Overlap, and Innovation*

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August 2026

Abstract

Minority stake acquisitions reduce innovation when the acquirer's and target's technologies overlap. Using worldwide patent, deal, and financial data for 1997–2020 and a staggered difference-in-differences design with propensity-score-matched controls, we find that five years after such an acquisition the cumulative patent output of acquirers and targets is 11.4% and 4.8% lower than that of comparable untreated firms. The decline is not confined to marginal, low-value filings; citation-weighted output falls in step with patent counts. It is concentrated in acquisitions with technological overlap—minority acquisitions between firms that do not share technology space show no comparable decline—and within treated firms both parties cut back in the classes they share with their counterpart, while targets' patenting outside the shared space rises relative to matched controls. The target response is strongest in acquisitions whose shared classes matter to both parties. The pattern points to reduced competitive incentives rather than financial constraints or the elimination of duplicative R&D. Most competition regimes do not scrutinize minority stake acquisitions, yet the cumulative innovation shortfalls we document are economically large and persistent.

Keywords: Minority Shareholdings, Innovation, Patents, Merger Policy

JEL: D22, G14, L13, L40, O31

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*We thank Chloé Le Coq, Pierre Fleckinger, Frédéric Jenny, Margaret Kyle, Joe Stiebale and Reinhilde Veugelers for thoughtful comments.

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1 Introduction

When a firm acquires a minority stake in another firm, antitrust authorities have traditionally looked the other way. The reason is that in virtually all jurisdictions, acquisitions that do not confer control are not subject to antitrust scrutiny, even when the buyer and target are competitors.[1] Yet these acquisitions are common,[2] and as we discuss in the next section, the literature has shown that they may soften competition, facilitate collusion, or lead to foreclosure. However, the effect of minority stake acquisitions on innovation is still largely underexplored, especially empirically. To fill this gap, we assemble worldwide patent, deal, and financial data over the period 1997–2020 and estimate the effect of minority stake acquisitions on the innovation activity of targets and acquirers.

From a theoretical standpoint, minority shareholdings can either strengthen or weaken the incentives of firms to innovate, depending on various factors such as whether the acquirer's innovation imposes a positive or a negative externality on the target and whether investments in innovation are strategic substitutes or strategic complements. In particular, once the acquirer obtains a stake in the target, it internalizes part of the externality that its innovation imposes on the target's profit. If the externality is negative—e.g., the firms are rivals—the acquirer would have a weaker incentive to innovate. To illustrate, suppose a firm acquires a 20 percent stake in a rival that patents in the same technology. Every dollar of profit that the firm's next innovation takes away from the rival now costs the acquirer 20 cents. The return to innovating against the rival falls. If the externality is positive—e.g., the firms are complementors or vertically related—the acquirer would have a stronger incentive to innovate. The target would respond in turn by either increasing or decreasing its innovation, depending on whether innovations are strategic substitutes or complements.

We focus on acquisitions where the pre-acquisition technologies of the acquirer and target directly overlap, in which case, the acquirer's technology is likely to impose a negative externality on the target. We measure technological overlap using 4-digit IPC (International Patent Classification) technology classes.[3] We classify acquisitions in which the acquirer and target share at least one 4-digit IPC class among patents applied for be-

[1] The European Commission's regulation, for instance, is limited to acquisitions that confer control and does not authorize the review of minority shareholdings. See Council Regulation (EC) No 139/2004, Article 3. In the U.S., Section 7 of the Clayton Act does cover partial stock acquisitions, but in practice very few minority deals are challenged (Rock and Rubinfeld, 2017).

[2] Calculations based on the Zephyr (Bureau van Dijk) database that we use in this paper show that minority stake acquisitions account for roughly 25% of global M&A activity.

[3] IPC is a global hierarchical system used to categorize patents into specific technical fields. There are over 650 distinct 4-digit IPC codes (e.g., G06F for "Data Processing"), which provide a granular measure of technological proximity.

fore the year of the acquisition and subsequently granted as "overlapping minority stake acquisitions."

Using a staggered difference-in-differences design with propensity-score-matched controls, we find that overlapping minority stake acquisitions change both how much and where the involved firms innovate. Five years after the acquisition, treated firms have accumulated significantly less patent output than their matched controls. The cumulative shortfall reaches 11.4% for acquirers and 4.8% for targets by year five. Citation-weighted patent output follows a similar decline, so the cutback is not confined to marginal, low-value patents. Point estimates are consistently larger for acquirers than for targets. These effects arise at ownership levels far below any control threshold. Most of the acquisitions we study leave the acquirer holding well under a quarter of the target, and all remain below majority control. We do not observe any such decline following non-overlapping minority stake acquisitions.

This paper makes three contributions. First, it provides the first large-scale causal evidence on how minority stake acquisitions affect innovation, based on worldwide deal, patent, and financial data. Second, it locates the effect in the technological relationship between the parties, with declines that arise only when the firms' technologies overlap and that grow with how much the shared space matters to both. Third, it speaks directly to the ongoing policy debate on partial-ownership review, identifying technological overlap as a screen for the transactions in which innovation incentives are most likely to change.

Acquirers and targets involved in overlapping minority stake acquisitions have, pre-acquisition, both overlapping and non-overlapping technologies. When we disaggregate to the level of individual technology classes within each firm, we find that, on net, the decline falls on the areas where the pre-acquisition technologies of the acquirer and target directly overlap. Moreover, both acquirers and targets cut innovation in overlapping technology classes. Outside the overlap, the two sides diverge. Targets' patenting in non-overlapping classes rises relative to matched controls, whereas acquirers reduce patenting in non-overlapping classes as well. These findings suggest that the post-acquisition decline in innovation reflects the internalization of externalities—once the acquirer holds a stake in a target, it cuts its investment in overlapping technologies and the target reacts to this decline by cutting its own innovation. The acquirer could in principle also use board seats or information rights to discourage the target's research. Non-overlapping acquisitions convey the same governance rights, however, and are not followed by any decline in patenting (Section 6.2).

The target response is concentrated in acquisitions in which the shared technology space matters to both parties, and likewise in acquisitions in which the two firms also overlap in product markets. Because our binary overlap definition counts any shared class, the average target effect understates the response of firms whose research programs genuinely collide. Alternative channels through which a minority stake acquisition can affect innovation, such as financial constraints or managerial overload, would affect all technology classes alike. This is difficult to reconcile with the targets' pattern. For acquirers, whose retreat is broader, the absence of any decline after non-overlapping acquisitions speaks against these channels.

The within-firm pattern also speaks against the elimination of duplicative R&D as the driving force. The most direct form of duplication elimination would have one of the two firms cut back while the other maintains its effort in the shared technology classes, most plausibly the target, since the acquirer bears only a fraction of the target's forgone profits but all of its own. Instead, both acquirers and targets cut their innovation in overlapping technology classes. Moreover, as we show in Section 6.4, acquirers cut back even in deals that offer little scope for duplication because the shared classes are marginal to one of the two research programs.

Recently, policymakers have started to be concerned with minority shareholdings. For instance, the 2023 FTC-DOJ Merger Guidelines now state that "cross-ownership and common ownership can reduce competition by softening firms' incentives to compete, even absent any specific anticompetitive act or intent." Moreover, the guidelines warn that a partial owner "may decide not to develop a new product feature to win market share from the firm in which it has acquired an interest, because doing so will reduce the value of its investment in its rival."[4]

The first prominent antitrust action against a minority shareholding came in 2015, when the UK Competition and Markets Authority required Ryanair to cut its 29.8% stake in Aer Lingus to 5%. In 2021, the DOJ limited Geisinger's planned 30% investment in a rival hospital to a 7.5% passive stake. The FTC has since opened an investigation into Microsoft's investment in OpenAI. And in 2025, the European Commission first fined Delivery Hero and Glovo € 329 million for cartel conduct facilitated by Delivery Hero's minority stake in Glovo, then required the divestiture of Prosus's 27.4% stake in Delivery Hero as a merger remedy, the first time it conditioned an approval on the sale of a

[4]Both quotations are from Guideline 11 of the U.S. Department of Justice and Federal Trade Commission (2023), the second from its Section 2; available at `https://www.justice.gov/atr/merger-guidelines/applying-merger-guidelines/guideline-11`.

minority holding.[5]

These cases, however, focus exclusively on price effects and coordination and do not consider the effects on innovation. This may not be surprising given that the effects of minority shareholding on innovation have received little empirical attention.

The rest of the paper is organized as follows. Section 2 discusses related literature. Section 3 presents our theoretical framework. Section 4 presents our data and Section 5 discusses the empirical strategy. Section 6 reports the results together with the robustness analyses. Section 7 concludes.

2 Related Literature

Early literature on minority stake acquisitions has shown that horizontal acquisitions can soften competition in the Cournot model (Reynolds and Snapp, 1986; Flath, 1991, 1992; Bolle and Güth, 1992; Reitman, 1994) or the Bertrand model (Shelegia and Spiegel, 2012), and can facilitate collusion in an infinitely repeated Cournot model (Malueg, 1992) or infinitely repeated Bertrand model (Gilo et al., 2006).[6] Moreover, minority stakes in a vertically related firm can lead to upstream and downstream foreclosure (e.g., Baumol and Ordover, 1994; Reiffen, 1998; Greenlee and Raskovich, 2006; Spiegel, 2013; Hunold and Stahl, 2016; Levy et al., 2018) and can also raise prices and harm consumers without foreclosure (Flath, 1989; Fiocco, 2016; Hunold and Schlütter, 2025).[7] These predictions have received some empirical support (e.g., Dietzenbacher et al., 2000; Brito et al., 2014; Nain and Wang, 2018; Heim et al., 2022).

A more recent literature has studied the effects of overlapping ownership (both cross and common ownership) on innovation. Most of this literature, however—e.g., López and Vives (2019) and Stenbacka and Van Moer (2023)—considers symmetric models in which all firms hold the exact same stake in one another and studies how a change in this symmetric stake affects innovation. In reality though, minority shareholdings are

[5]Case AT.40700 (European Commission, 2025) and Case COMP/M.11936 (Paul, Weiss, Rifkind, Wharton & Garrison LLP, 2025). For the other cases in this and the preceding sentences, see `https://www.gov.uk/cma-cases/ryanair-aer-lingus-merger-inquiry` (Ryanair/Aer Lingus), `https://www.fiercehealthcare.com/hospitals/doj-reaches-settlement-geisinger-health-evangelical-antitrust-case` (Geisinger), `https://www.nytimes.com/2024/01/26/business/dealbook/ftc-ai-deals-microsoft-openai.html` (Microsoft/OpenAI), and `https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1356` (Delivery Hero/Glovo).

[6]Malueg (1992) shows that cross ownership can also hinder collusion in a repeated Cournot model, but when it does, firms should have no incentives to acquire ownership stakes in one another in the first place.

[7]Vertical minority stakes can also have pro-competitive effects. For instance, Flath (1989) shows that a minority stake in a downstream buyer can relax the double marginalization problem and thereby lower prices and benefit consumers. By contrast, a minority stake in an upstream supplier may have the opposite effect.

asymmetric and typically firms unilaterally acquire stakes in targets. Indeed, in our data, we observe cases where only one firm acquires a minority stake in another firm rather than all firms doing so. Shelegia and Spiegel (2024) consider a model where firms possibly hold asymmetric stakes in one another. They show that post-acquisition, the acquirer invests less in R&D whereas the target invests more. In our data both parties reduce their innovation, which within this class of models corresponds to the case in which R&D investments are strategic complements (Section 3). The overall effect on consumers is either positive or negative, depending on various factors such as the size of the ownership stakes, how symmetric they are, the size of the innovation, its marginal cost, and whether innovation is drastic (the innovating firm becomes a monopoly) or not.[8]

There is a related literature that studies how common ownership by institutional investors affects innovation. For instance, Antón et al. (2023) use data on patent citations of publicly listed U.S. corporations and show that an increase in common ownership is associated with a decrease in citation-weighted patents when products are sufficiently close substitutes, but an increase in citation-weighted patents when technology spillovers are relatively large. By contrast, Lewellen and Lowry (2021) study the effects of substantial increases in common ownership due to mergers of financial institutions, but find no significant effects on either firm profitability or firm R&D.[9] While related, common ownership is in general distinct from cross ownership. Under the latter, the acquirer internalizes an externality that its actions impose on the target and adjusts its behavior accordingly; the target in turn reacts to this adjustment. Under common ownership by contrast, the common owners internalize the externalities that the various firms they own impose on each other and possibly affect the behavior of these firms to maximize their own wealth.

Our paper is also related to the literature that studies the effects of horizontal mergers on investments in innovation; see Lefouili and Madio (2026) for a recent review of the relevant literature. Minority shareholding can be viewed as a "partial merger," but does not involve direct control and hence is distinct from full mergers. In particular, unlike a full merger, where the acquirer controls the target's strategy, in a minority stake acquisition, the acquirer can only affect its own strategy and the target only reacts to the change in the acquirer's strategy.

The closest study to ours is Bostan and Spatareanu (2018). They provide empirical evidence of increased innovation following minority acquisitions accompanied by cash

[8]Bayona and López (2018) and Ghosh and Morita (2017) are also related.

[9]See also (e.g., Antón et al., 2024; Li et al., 2023; Lindsey, 2008; Kostovetsky and Manconi, 2020; Borochin et al., 2020).

flows. The mechanism they focus on, however, is financial. The effect is present only when the acquisition is accompanied by cash flows to small, young, most financially constrained target firms with relatively small patent portfolios. There is no similar effect in minority acquisitions which do not involve cash transfers to target firms, or if the minority shares are acquired from a priori large patent portfolio firms, or the target is not financially constrained.[10] By contrast, we find that the decline is tied to technological overlap. It appears only after acquisitions between firms whose technologies overlap, and both parties cut back in the classes they share. This suggests that the driving force in our data is due to the internalization of externalities rather than financial considerations.

3 Theory

We develop a theoretical framework that establishes the conditions under which an increase in the stake of one firm in another will strengthen or weaken the incentives of the acquirer and target to invest in R&D. As we will show below, the conditions depend critically on two factors: whether the acquirer's R&D investment imposes a positive or a negative externality on the target and whether the R&D investments of the acquirer and target are strategic substitutes or strategic complements.[11]

Consider a duopoly where one firm (an acquirer) acquires a minority stake α in the other firm (the target firm). Let x^a and x^t be the R&D investments of the acquirer and target and let their corresponding profit functions be $\pi^a(x^a, x^t)$ and $\pi^t(x^a, x^t)$. Assuming that the acquisition does not involve control, each firm chooses its investment in R&D to maximize its overall profit. The acquirer chooses x^a to maximize

$$\Pi^a(x^a, x^t, \alpha) = \pi^a(x^a, x^t) + \alpha\pi^t(x^a, x^t), \tag{1}$$

and the target chooses x^t to maximize

$$\pi^t(x^a, x^t). \tag{2}$$

[10] Bena and Li (2014) investigate which characteristics of corporate innovation activities determine whether a firm becomes an acquirer or a target firm. They show that firms with large patent portfolios and low R&D expenses are acquirers, while firms with high R&D expenses and slow growth in patent output are more likely to be targets. They also show that acquirers with prior technological linkage to their targets produce more patents after the acquisition.

[11] Aoki and Spiegel (2009) show that when R&D competition is followed by competition in the product market, R&D investments may be either strategic substitutes or strategic complements, depending on whether it is more important for firms to take the technological lead or to catch up in case another firm takes the technological lead.

A Nash equilibrium in the R&D game, $(\hat{x}^a, \hat{x}^t)$, is given by the solution to the following first-order conditions:

$$\Pi_a^a(x^a, x^t, \alpha) = \pi_a^a(x^a, x^t) + \alpha \pi_a^t(x^a, x^t) = 0, \qquad \text{and} \qquad \pi_t^t(x^a, x^t) = 0, \tag{3}$$

where subscripts denote partial derivatives. The expression $\pi_a^t(x^a, x^t)$ represents the externality that the acquirer's R&D investment imposes on the target. In general, the acquirer's R&D investment boosts the acquirer's sales. If the target sells a complementary product or is a trading partner of the acquirer (either a customer or a supplier), it benefits from the increase in the acquirer's sales, i.e., $\pi_a^t(x^a, x^t) > 0$. By contrast, if the target is a rival, the increase in the acquirer's sales harms the target, i.e., $\pi_a^t(x^a, x^t) < 0$.

We will assume that the second-order conditions are satisfied, i.e., $\Pi_{aa}^a(x^a, x^t, \alpha) < 0$ and $\pi_{tt}^t(x^a, x^t) < 0$, and will assume in addition that the following inequalities, which ensure that the Nash equilibrium is unique and stable (see Shapiro, 1989), hold:

$$\Pi_{aa}^a(x^a, x^t, \alpha) + \left| \Pi_{at}^a(x^a, x^t, \alpha) \right| < 0, \qquad \text{and} \qquad \pi_{tt}^t(x^a, x^t) + \left| \pi_{at}^t(x^a, x^t) \right| < 0. \tag{4}$$

The first-order conditions implicitly define the best-response functions of the two firms. Using the first-order conditions, the slopes of the best-response functions in the (x^a, x^t) space are given by:

$$\frac{\partial x^a}{\partial x^t} = -\frac{\Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha)}{\Pi_{aa}^a(\hat{x}^a, \hat{x}^t, \alpha)}, \qquad \text{and} \qquad \frac{\partial x^t}{\partial x^a} = -\frac{\pi_{at}^t(\hat{x}^a, \hat{x}^t)}{\pi_{tt}^t(\hat{x}^a, \hat{x}^t)}. \tag{5}$$

The strategies of the acquirer and target are strategic substitutes if the slopes are negative and are strategic complements if they are positive. Given the second-order conditions, the strategies of the acquirer and target are strategic substitutes if $\Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha) < 0$ and $\pi_{at}^t(\hat{x}^a, \hat{x}^t) < 0$, and are strategic complements if $\Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha) > 0$ and $\pi_{at}^t(\hat{x}^a, \hat{x}^t) > 0$.

We now examine how the acquirer's stake in the target firm, α, affects the R&D investments. To this end, we fully differentiate the first-order conditions and divide through by $\partial \alpha$. The resulting comparative matrix is given by

$$\begin{pmatrix} \Pi_{aa}^a(\hat{x}^a, \hat{x}^t, \alpha) & \Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha) \\ \pi_{at}^t(\hat{x}^a, \hat{x}^t) & \pi_{tt}^t(\hat{x}^a, \hat{x}^t) \end{pmatrix} \begin{pmatrix} \frac{\partial \hat{x}^a}{\partial \alpha} \\ \frac{\partial \hat{x}^t}{\partial \alpha} \end{pmatrix} = \begin{pmatrix} -\pi_a^t(x^a, x^t) \\ 0 \end{pmatrix}. \tag{6}$$

Solving (6), yields

$$\frac{\partial \hat{x}^a}{\partial \alpha} = -\frac{\pi^t_a(x^a, x^t) \times \pi^t_{tt}(x^a, x^t)}{\Delta}, \qquad \text{and} \qquad \frac{\partial \hat{x}^t}{\partial \alpha} = \frac{\pi^t_a(x^a, x^t) \times \pi^t_{at}(x^a, x^t)}{\Delta}, \tag{7}$$

where $\Delta = \Pi^a_{aa}(\hat{x}^a, \hat{x}^t, \alpha)\pi^t_{tt}(\hat{x}^a, \hat{x}^t) - \Pi^a_{at}(\hat{x}^a, \hat{x}^t, \alpha)\pi^t_{at}(\hat{x}^a, \hat{x}^t)$ is the determinant of the comparative statics matrix. By (4), $\Delta > 0$.[12]

Equation (7) has three implications. First, an increase in α does not affect the R&D investments of either the acquirer or the target if $\pi^t_a(x^a, x^t) = 0$, i.e., the acquirer's R&D investment does not impose an externality on the target. Second, given that $\pi^t_{tt}(x^a, x^t) < 0$, an increase in α boosts the acquirer's R&D investment if $\pi^t_a(x^a, x^t) > 0$, i.e., the externality is positive, but decreases it if the externality is negative, i.e., $\pi^t_a(x^a, x^t) < 0$. Third, the target reacts in the same direction as the acquirer if R&D investments are strategic complements, i.e., $\pi^t_{at}(x^a, x^t) > 0$, and reacts in the opposite direction if strategies are strategic substitutes, i.e., $\pi^t_{at}(x^a, x^t) < 0$.

Equation (7) thus yields four predictions for our setting. First, an acquirer facing a negative externality reduces its R&D investment (P1). Second, the target moves in the same direction if R&D investments are strategic complements (P2). Third, because the target's reaction operates through the slope of its best-response function, which is smaller than one in absolute value by condition (4), the target's response is smaller in absolute value than the acquirer's (P3). Fourth, absent an externality—in our setting, absent technological overlap—neither side responds (P4).

4 Data

4.1 Data Sources

We measure innovation using PATSTAT, maintained by the European Patent Office, which covers worldwide patent applications from 1844 to 2023. We extract patent application IDs, application years, forward citations, and 4-digit IPC technology classes for all granted patents, dated by application year. This gives us two proxies for innovation output, patent quantity (number of grants) and patent quality (number of forward citations). Our M&A data come from Bureau van Dijk's Zephyr database, which records deal-level information—including completion dates, stake percentages, and acquirer and target identities—for transactions worldwide from 1997 to 2020. We merge both datasets

[12] $\Delta > 0$ ensures that the Nash equilibrium is stable in the sense that the best-response function of the acquirer crosses the best-response function of the target firm in the (x^a, x^t) space from above if strategies are strategic substitutes and cuts it from below if the strategies are strategic complements.

with Bureau van Dijk's Orbis database, which tracks balance sheet and P&L information for approximately 45 million firms worldwide. The merged dataset provides, for each firm-year, patent counts, citation counts, the list of IPC classes in which the firm is active, and financial variables including sales, total assets, liquidity ratios, and employment.

4.2 Identifying Overlapping Minority Stake Acquisitions

From the universe of Zephyr transactions, we select completed deals where the reported final stake is below 50%. We exclude share buybacks and self-tenders, deals where either party is classified by primary SIC code as an Investor (SIC code 6799) or a Management investment office (SIC code 6722), and transactions where patent data are unavailable for both the acquirer (or its immediate parent) and the target. In roughly 17% of all minority stake deals, both the acquirer and the target hold at least one patent. This is a remarkably high share, as only about 5% of firms in Orbis hold any patent at all (Gugler et al., 2023), so if patenting were unrelated to deal participation, the share of deals in which *both* parties patent would be well below 1%. Firms involved in minority stake acquisitions are thus far more technology-intensive than the typical firm. This leaves 24,267 minority stake acquisitions.

As we show in Section 3, a minority stake acquisition is expected to affect innovation incentives only if the acquirer's innovation imposes an externality on the target. We expect to observe an externality only if the pre-acquisition technologies of the acquirer and target are overlapping. Therefore, a key classification step is identifying which minority stake acquisitions involve firms with overlapping technologies. Specifically, we classify a deal as an "overlapping minority stake acquisition," if the acquirer (or its immediate parent) and target share at least one 4-digit IPC class among patents applied for before the year of the acquisition and subsequently granted.[13] With more than 650 distinct 4-digit IPC classes, this approach provides a fine-grained measure of technological overlap.[14] By this definition, 56% of the 24,267 minority stake acquisitions involve overlapping technologies, corresponding to 4,561 unique targets and 2,898 unique acquirers.

[13]We use the immediate parent rather than the ultimate corporate owner, since ultimate owners are often diversified conglomerates whose broad patent portfolios would generate spurious overlapping classifications.

[14]Statistics based on World Intellectual Property Organization, available at `https://www.wipo.int/classifications/ipc/en/ITsupport/Version20240101/transformations/stats.html`.

4.3 Treated Sample

We analyze targets and acquirers separately, constructing two parallel datasets. The details of how we constructed the datasets are summarized in Table A1 in the Appendix. Starting from the full set of deals which involve overlapping technologies, we first isolate the effect of minority stakes from a subsequent change of control by excluding deals in which the acquirer obtains majority control of the target within five years of the minority acquisition.[15] We then drop observations without Orbis financial data, leaving 2,783 targets and 1,311 acquirers, and further remove non-corporate entities (financial firms, mutual funds, pension funds) and firms without at least three consecutive years of patent data. Thus, 2,682 targets and 1,239 acquirers enter the matching procedure described in Section 5. We discuss the representativeness of this sample in Appendix A.1.

4.4 Descriptive Statistics

The treated firms in our final sample are headquartered in 37 countries. Most deals in our sample are domestic.[16] Japan accounts for the largest share of overlapping minority stake acquisitions (34%), followed by South Korea (19%)—a prominence that reflects the Japanese *keiretsu* and the Korean *chaebol*, business groups whose member firms are tied together by extensive webs of reciprocal minority shareholdings, typically held for decades as a device for relational governance and group cohesion. We return to the role of these two countries in the robustness analysis in Section 6. The sample is concentrated in manufacturing (68%) and information industries (6%); detailed breakdowns by sector appear in the Appendix.

The distribution of acquired stakes is right-skewed, with most observations below 25% (see Figure A.4 in the Appendix).[17] Almost half of the sample firms (48%) are involved in more than one overlapping minority acquisition during the observation period, sometimes as acquirer, sometimes as target.[18]

Table A3 in the Appendix provides pre-acquisition summary statistics. Acquirers

[15] This exclusion conditions the sample on a post-acquisition event. We show in Section 6.6 that the results are essentially unchanged when these deals are retained and the affected firms are instead censored from the year the stake turns into a majority position.

[16] 68% of targets were acquired by firms in the same country; 80% of acquirers acquired domestic targets. Detailed breakdowns by year, country, and industry appear in Figures A.1–A.3 in the Appendix.

[17] The distribution differs between the acquirer and target samples because some acquisitions are reported in one sample but not in the other due to data availability. For instance, financial or patent data might be available for the acquirer but not the target or vice versa.

[18] Approximately 28% of the targets were subject to successive stake acquisitions or separate transactions over time by the same acquirer; about 17% of acquirers acquired stakes in the same target more than once. In addition, roughly 22% of sample firms were also involved in non-overlapping minority stake acquisitions.

tend to be substantially larger than targets. At the median, a target has about one tenth of the sales and total assets of its acquirer, and an acquirer roughly four times the sales and total assets of its target (Table A4).[19] Innovation activity is highly concentrated, with the top 15% of firms in our sample accounting for more than 90% of the total patent stock, consistent with the well-documented concentration of global R&D activity.[20]

5 Research Design

Acquirers are larger, more innovative, and more liquid than other firms (see the pre-acquisition summary statistics in Table A3 in the Appendix). Acquisitions also occur at different times. Our empirical strategy will therefore involve a combination of propensity score matching with a staggered difference-in-differences estimator.

5.1 Matching Procedure

Potential controls are corporate Orbis firms with sufficient patent and financial information (at least five consecutive years of patent data) that were never involved in a majority acquisition or a minority stake acquisition. This leaves more than 92,000 firms, more than 40 times the number of treated firms, so that a sufficiently close match can be found for each treated observation.

From this pool we select controls by propensity score matching (Rosenbaum and Rubin, 1983, 1985). A probit model predicts the probability of being involved in an overlapping minority stake acquisition based on pre-treatment observables, namely sales, employment, total assets, intangible assets, liquidity ratio, and firm age, all measured at $t-1$. It also includes the levels of cumulative patents and cumulative citations at $t-1$, $t-2$, and $t-3$, so that treated and control firms are similar in their recent pre-acquisition patent and citation histories. The model includes country, year, and industry (2-digit NAICS) fixed effects. Table A2 in the Appendix reports the estimates. The pseudo-R^2 is 0.41 for targets and 0.45 for acquirers, confirming that treated firms differ substantially from the general population along observable dimensions.

Each treated firm is matched to the control firm with the closest propensity score, sub-

[19]The two panels condition on different subsets of deals—those for which the counterpart's financial data are available in our panel—which is why the two median ratios are not exact inverses of each other.

[20]Based on the 2022 EU Industrial R&D Investment Scoreboard, the top 15% of the 2,500 largest R&D investors account for nearly 75% of total Scoreboard R&D expenditure—and this within a group that is already restricted to the world's largest R&D spenders. Measured over a broader population such as ours, concentration is naturally even higher. See `https://iri.jrc.ec.europa.eu/scoreboard/2022-eu-industrial-rd-investment-scoreboard`.

ject to three restrictions. First, treated and control firms must share the same observation year. Second, potential controls must hold at least one patent in a 4-digit IPC class in which the treated firm also patented prior to the acquisition. This ensures that matched pairs operate in the same technology domain.[21] Together with the propensity-score covariates, this requirement makes treated and control firms subject to similar conditions in their technology fields. Third, we enforce common support by dropping the few treated firms (12 targets and 11 acquirers) whose propensity score exceeds the maximum among potential controls, implying that they do not have a sufficiently similar control firm.

We match with replacement, so a control firm can serve as a match for more than one treated firm; in the estimation, each control firm is weighted by the number of treated firms it is matched to. Pairs whose propensity scores differ by more than 0.05 in absolute terms are dropped. We additionally require treated and control firms to have similar pre-acquisition growth in cumulative patent and citation output, with a maximum absolute difference of 0.1 in their five-year log slopes, so that matched pairs are on comparable innovation trajectories before the acquisition. Of the 2,682 targets and 1,239 acquirers that enter the procedure, 2,205 and 1,033 have complete pre-treatment covariates, and 2,069 targets and 919 acquirers find a match, drawing on 1,304 and 666 distinct control firms.

After matching, the standardized biases of the innovation stocks—the covariates most critical for our outcomes—are essentially eliminated, and the biases of the remaining covariates fall by more than 90% (Figure A.5 in the Appendix). The pseudo-R^2 of the probit model re-estimated on the matched sample drops from 0.41–0.45 to 0.05–0.08. Matched treated and control firms are also much more similar in their patent distributions than unmatched firms (Figures A.6 and A.7 in the Appendix).

5.2 Staggered Difference-in-Differences

Different acquisitions occur at different times, so the standard two-way fixed effects (TWFE) estimator may produce biased estimates in the presence of heterogeneous or dynamic treatment effects. We use the de Chaisemartin and D'Haultfoeuille (2026) estimator, which in any given period compares newly treated units against units that remain untreated at that point. This avoids implicitly comparing newly treated units against already-treated units and accommodates heterogeneous treatment timing, dynamic ef-

[21] A potential concern is that control firms sharing 4-digit IPC classes with treated firms may themselves be affected by the minority stake acquisition once it affects the R&D investments of the acquirer and target. We address this concern in Section 6.6 by re-matching treated firms to controls that share only 3-digit IPC overlap but have no 4-digit overlap.

fects, and varying treatment intensities—all features of our setting, where acquisitions of different sizes occur at different points in time.[22]

The dependent variable is the cumulative patent output (or cumulative forward citations) of firm i in year t, expressed as $\ln(\text{Patents}_{i,t} + 1)$ or $\ln(\text{Citations}_{i,t} + 1)$.[23] Cumulative patent output captures the innovative output a firm has built up and hence the resulting difference in its technological position relative to the counterfactual.[24] At each event horizon, the coefficient measures the cumulative output gap between treated firms and their matched controls. Annual patent counts are volatile at the firm level—for the median firm, the standard deviation of the annual count is as large as its mean—and the cumulative outcome averages over this volatility. The pattern is not specific to the cumulative representation. Annual patent production, scaled by the firm's pre-acquisition patent stock, shows the same decline (Section 6).

Let $\tau \in \{-5, \ldots, 5\}$ denote event time relative to the last year before the firm's first overlapping minority stake acquisition, so that $\tau = 0$ is this last pre-acquisition year and serves as the reference period, $\tau = 1$ is the acquisition year, and $\tau = 5$ is the fifth year of exposure. The event-study specification is:

$$\ln(\text{Patent}_{i,t} + 1) \; = \; \gamma_i + \gamma_t \; + \sum_{\substack{\tau=-5, \\ \tau \neq 0}}^{5} \beta_\tau \, D_{i,t}^{\tau} \; + \; u_{i,t} \tag{8}$$

where $D_{i,t}^{\tau} = 1$ if year t lies τ years from firm i's reference period. The specification includes firm fixed effects γ_i and calendar-year fixed effects γ_t, and standard errors are clustered at the firm level. Equation (8) fixes the notation and the timing convention used in all event-study figures; the β_τ themselves are not estimated by least squares but by the de Chaisemartin and D'Haultfoeuille (2026) estimator described above, which recovers each β_τ from comparisons of newly treated firms with firms that are not yet treated at that point. We consider a post-treatment window of five years; the sample thins at longer lags. Our primary estimand is the cumulative output gap five years after the acquisition. The event-study coefficients report this gap at each horizon and, for the pre-acquisition years, provide a direct check on the comparability of treated firms and their matched controls.

[22]See, e.g., de Chaisemartin and D'Haultfoeuille (2026), de Chaisemartin and D'Haultfoeuille (2020), Borusyak et al. (2024), Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Goodman-Bacon (2021).

[23]We count cumulative granted patents, regardless of whether they already expired or were not renewed. We use $\ln(x + 1)$ for patents and citations for consistency across outcomes. Results for cumulative patents are similar using $\ln(x)$, with year-five effects of -12.7% for acquirers and -5.3% for targets.

[24]Cumulative and fixed multi-year patent outcomes are also used by Bena and Li (2014), Coelli et al. (2022), and Xie and Zhou (2026).

6 Results

6.1 Main Effects on Patent Output

We begin with the event-study estimates from the staggered difference-in-differences design described in Section 5. Figure 1 reports the cumulative output gap at each event horizon, together with the pre-acquisition comparisons, for acquirers (left panel) and targets (right panel). The event-study coefficients are anchored at the firm's first overlapping acquisition and are identical under the three treatment codings of Table 1, which differ only in how the average effect is scaled.

In the matched sample, the estimated pre-acquisition coefficients are small and jointly statistically indistinguishable from zero; a joint test that all five placebo coefficients are zero has p-values of 0.86 for targets and 0.19 for acquirers. Because pre-acquisition patent histories enter the matching procedure, these coefficients should be interpreted as describing the matched sample rather than as an independent validation of the matching design. The annual specification of Figure A.18, which does not cumulate, shows equally small pre-acquisition coefficients. Section 6.6 probes the sensitivity of the estimates to remaining pre-trend concerns, to the role of individual countries, and to further threats to identification. The post-acquisition estimates are negative. We estimate that by year five treated acquirers have accumulated 11.4% less patent output, and treated targets 4.8% less, than their matched controls.[25]

The decline also appears directly in annual patent production. When we estimate the same design on yearly granted patents scaled by the firm's pre-acquisition patent stock, annual output is lower throughout the post-acquisition window, reaching 7.0 patents per 100 pre-acquisition patents for acquirers and 6.9 for targets at year five, with small pre-acquisition coefficients (Figure A.18 in the Appendix). As with the cumulative outcome, non-overlapping acquisitions show no comparable decline. The cumulative gap thus reflects lower annual patent production after the acquisition.

[25] These estimates are robust to standard variations of the matching and weighting procedure and to the treatment definition. Estimating the model without weighting control firms by how often they serve as a match yields year-five effects of −14.3% for acquirers and −7.5% for targets, matching without replacement yields −12.2% and −6.0%, and matching each treated firm to its three nearest neighbors instead of one yields −11.1% and −5.7%. The corresponding event studies for patents are shown in Figures A.10, A.11, and A.12 in the Appendix. Restricting exposure to the first acquisition, by censoring treated firms at their second overlapping acquisition, yields −11.6% and −3.4% (Figure A.15 in the Appendix).

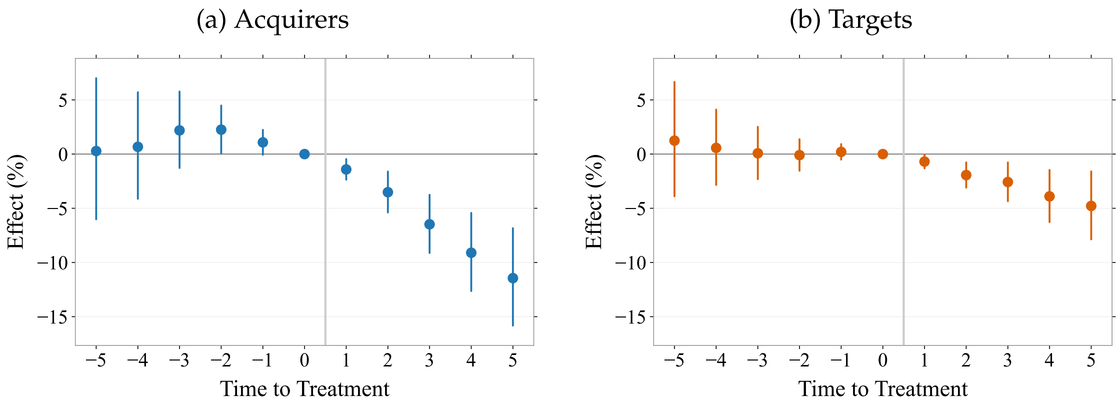

Figure 1: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. The event-study coefficients are invariant to the treatment coding of Table 1 (binary, discrete, or continuous). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

Table 1 complements the event-study figures by reporting the average effect of treatment over the full five-year post-acquisition window. Because the average includes the early post-treatment years, when the cumulative gap is still small, the magnitudes are smaller than the year-five endpoints.

Panel A uses a binary treatment indicator; here we find an average decline in cumulative patent counts of 5.7% for acquirers and 2.6% for targets. Panel B codes the treatment as a count of overlapping minority acquisitions per firm. That is, 0 before the first acquisition, 1 after the first acquisition, 2 after the second acquisition, etc. The pattern is similar to Panel A, with slightly smaller point estimates.

Panel C treats the cumulative acquired ownership stake as a continuous treatment, so that the effect is expressed per percentage point of ownership rather than per acquisition. This puts acquisitions of different size on a common scale. Under the assumption that a firm's response is proportional to the size of the stake, an additional percentage point of acquired ownership corresponds to a decline of roughly 0.3–0.4% in cumulative patenting. Whether the response is in fact proportional is a question our data cannot fully resolve—minority stakes are below 50% by construction and most are well below 25%. Splits by stake size are too coarse to pin down the functional form, but they show that the effects do not depend on the largest stakes.[26]

[26]Restricting the treated sample to acquisitions with an initial stake below 20 percent, which covers 89

Across all specifications, point estimates are larger for acquirers than for targets. This ordering is compatible with prediction P3 of Section 3, but the relative cumulative coefficients are not a direct test of the relative adjustment in R&D effort.

Table 1: Estimated Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output

	Acquirers	Targets
	(1)	(2)
Panel A: *Binary treatment*		
Treatment effect	-0.059^{***}	-0.026^{***}
	(0.012)	(0.008)
Panel B: *Discrete treatment*		
Treatment effect	-0.052^{***}	-0.020^{***}
	(0.011)	(0.007)
Panel C: *Continuous treatment*		
Treatment effect	-0.004^{***}	-0.003^{***}
	(0.001)	(0.001)
Observations	24,038	57,927

Each column reports the estimated average effect over the five-year period post-acquisition from a staggered difference-in-differences regression using the de Chaisemartin and D'Haultfoeuille (2026) estimator. The results represent weighted averages of group-time effects. Panel A treats the acquisition as binary (0 before an overlapping minority stake acquisition, 1 thereafter). Panel B treats the treatment as discrete (0 before, 1 after the first deal, 2 after the second, etc.). Panel C codes the treatment as the cumulative percentage stake acquired in a rival, so that the coefficients are expressed per percentage point of acquired ownership. Mechanically this normalizes the average effect of Panel A by the average stake held over the post-acquisition window, which is 16.8% for acquirers and 9.6% for targets; it is a per-percentage-point effect under the proportionality assumption discussed in the text. The event-study figures report effects in percent, $100 \times (\exp(\beta) - 1)$, whereas the table reports log-point coefficients. All specifications include firm and year fixed effects and use a 5-year post-acquisition window. Standard errors are clustered at the firm level and shown in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Does Technological Overlap Matter?

Having established that innovation declines after overlapping minority stake acquisitions, we now turn to what drives the decline. If it is driven by factors such as managerial

percent of treated targets and 67 percent of treated acquirers, leaves the year-five effects essentially unchanged at -11.4% for acquirers and -4.8% for targets, with small pre-acquisition coefficients (Figure A.17 in the Appendix). The target response survives much tighter caps and remains significant below 10 percent and even below 5 percent, ownership levels at which any influence over the counterpart is implausible. The acquirer point estimate below 10 percent falls to -5.2% and loses precision, consistent with the acquirer internalizing the externality in proportion to its stake.

overload, governance costs, or financial frictions associated with the acquisition, rather than by the internalization of externalities, we would expect to observe it following all acquisitions, including non-overlapping ones. To examine whether this is the case, we run our analysis on non-overlapping minority stake acquisitions, defined as acquisitions where, pre-acquisition, the acquirer and target do not share any 4-digit IPC class. Mirroring the treatment definition of the main analysis, treatment is assigned at the firm's first minority stake acquisition; firms that also experience an overlapping acquisition in the acquisition year or the following five years are excluded, so that the placebo estimate is not contaminated by overlapping deals. For this exercise we repeat the full matching procedure for the non-overlapping treated firms, again drawing controls from firms that were never party to a minority stake acquisition or a majority acquisition. Figure 2 plots the event study. The coefficients are small throughout the post-acquisition window and show none of the cumulative-gap pattern of Figure 1; no post-acquisition coefficient is statistically significant for either side. Table 2 confirms that the average treatment effects are -0.2% for acquirers and -0.8% for targets, neither statistically significant. We find no comparable decline following minority stake acquisitions between firms without pre-acquisition technological overlap.

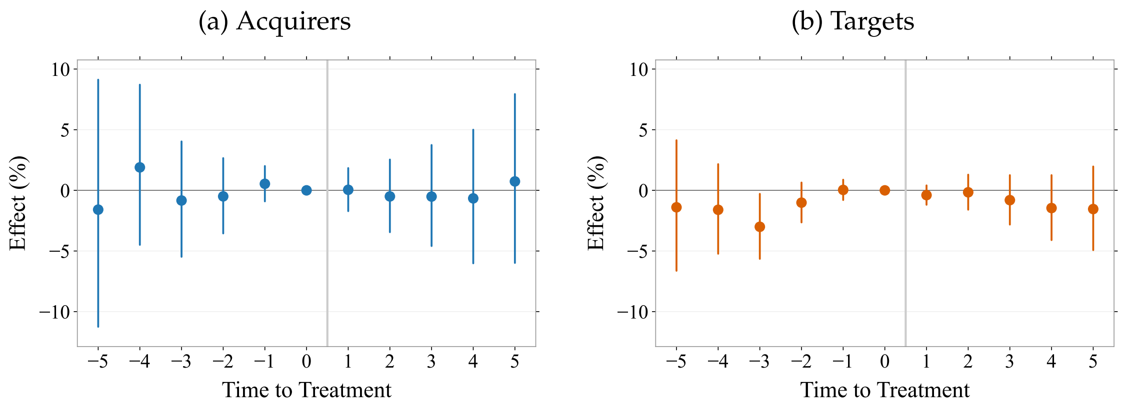

Figure 2: Event Study on the Effects of Non-Overlapping Minority Stake Acquisitions on Cumulative Patent Output

The figure presents coefficient estimates in percentage for each of the five years before and five years after the non-overlapping minority stake acquisition of treated firms. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

Table 2: Estimated Effects of Non-Overlapping Minority Stake Acquisitions on Cumulative Patent Output

	Acquirers	Targets
	(1)	(2)
Treatment effect	−0.002	−0.008
	(0.019)	(0.009)
Observations	10,141	34,822

Each column reports the estimated average effect over the five-year period post-acquisition from a staggered difference-in-differences regression using the de Chaisemartin and D'Haultfoeuille (2026) estimator. The results represent weighted averages of group-time effects. The acquisition is treated as binary (0 before a non-overlapping minority stake deal, 1 thereafter). All specifications include firm and year fixed effects and use a 5-year post-acquisition window. Standard errors are clustered at the firm level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Innovation Inside and Outside the Shared Technology Space

Overlapping minority stake acquisitions reduce patenting at the firm level. This finding alone does not reveal the underlying mechanism. A general disruption to the firm's operations would depress patenting uniformly across its entire technology portfolio, including in fields the firm enters only after the acquisition. A change in competitive innovation incentives, by contrast, should be visible specifically in the firms' shared technology space.[27]

To distinguish between these explanations, we exploit variation *within* each firm's technology portfolio. For each acquisition, we classify a firm's technology areas into *overlapping* classes—those 4-digit IPC classes where both the acquirer and target held at least one patent prior to the acquisition—and *non-overlapping* classes, where they do not. The classification errs on the side of caution. Since 4-digit IPC classes are narrow, some technologically related classes end up in the non-overlapping group, biasing against finding a difference. We then collapse each firm's patenting into two components—the cumulative number of patents in overlapping classes and in non-overlapping classes[28]—and estimate the event study of equation (8) separately for each component, using the same estimator and treatment definition as in the firm-level analysis. The overlapping component is measured on the same set of classes for both firms, namely the treated firm's overlapping classes, which are fixed before the acquisition. Because the control firm con-

[27] Non-overlapping technology classes may also be affected directly if they are complements or substitutes to the overlapping technologies.

[28] Patents often carry several IPC classes, and a patent counts toward each class it is assigned to, so the two components need not sum to the firm-level total. Among the patents entering the non-overlapping component, 5.6% for targets and 15.6% for acquirers also carry one of the overlapping classes.

tributes its patenting in these very classes, a shared field that loses momentum for reasons unrelated to the acquisition slows both series alike and differences out of the comparison. The non-overlapping component covers each firm's patenting in all other classes.[29]

Figure 3 shows the results. In the overlapping classes, the year-five cumulative patent-output gap reaches -12.6% for acquirers and -6.6% for targets—magnitudes larger than the firm-level effects. Outside the shared classes, the two sides differ. For targets, cumulative patenting in non-overlapping classes is 8.3% higher than for matched controls by year five. Acquirers instead show a gap on both margins, inside and outside the shared classes, with outside patenting 7.2% lower by year five. The rise outside the shared space does not offset the targets' retreat, as total patent output and citation-weighted output both fall (Table 1 and Section 6.5). It is also noteworthy that targets—which do not internalize any part of the acquirer's profit—cut their overlapping-class patenting substantially as well. A pure elimination-of-duplication motive on the acquirer's side would leave the target's incentives unchanged; the pattern is instead consistent with both firms softening R&D competition in the shared technology space. Section 6.4 examines how these responses vary with how much the shared technology space matters to the two parties.

[29]For this analysis we additionally require control firms to have a similar pre-acquisition composition of their patent portfolio. The share of a control firm's pre-acquisition patenting that falls into its treated counterpart's overlapping classes must be within 10 percentage points of the treated firm's own share.

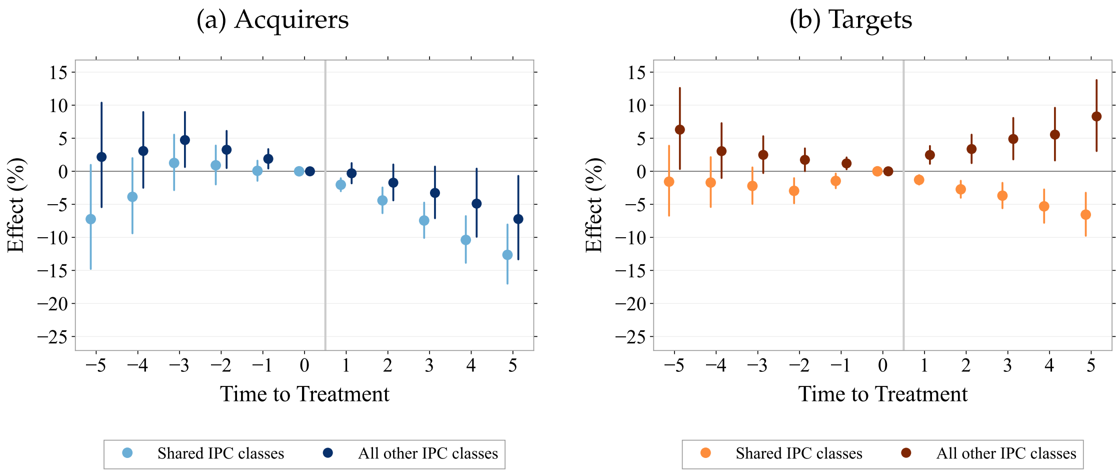

Figure 3: Event Studies for Shared and All Other Technology Classes

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms, estimated separately for the firm's shared IPC classes (light circles) and all of its other IPC classes (dark squares). The dependent variable is the log of the cumulative number of granted patents (dated by application year) in the respective set of classes, per firm and year. The overlapping component is measured on the same pre-acquisition set of overlapping IPC classes for the treated firm and its matched control; the non-overlapping component comprises each firm's own remaining IPC classes. The estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each series shows 95% confidence intervals based on standard errors clustered at the firm level.

The pattern ties the decline to the shared technology space. This is consistent with the internalization mechanism. Once the acquirer holds a stake in its rival, the return to competing along the shared technological trajectory falls. It also distinguishes the mechanism from firm-wide explanations such as managerial distraction or financial frictions. For targets, the contrast between the two components makes this direct. A firm short of attention or funding would scale back across its portfolio, not cut back in the classes it shares with its counterpart while expanding elsewhere. For acquirers, whose retreat is broader, the same distinction is provided by the placebo of Section 6.2. Acquisitions without technological overlap carry the same governance and integration frictions, yet leave patenting unchanged. At the firm level, the net effect is the moderate aggregate decline documented in Table 1, which masks a larger contraction in the classes the two firms had in common before the acquisition.

The decline in the shared classes could in principle reflect a change in patenting rather than in innovation. Once the rival holds a stake, patents filed mainly to block that rival or to deter litigation lose part of their purpose, and firms may protect more of their work through secrecy. Citation-weighted output, however, falls at least as strongly as patent counts (Section 6.5), and a lower propensity to patent in the contested classes would not produce the targets' expansion outside them. Nor is cooperation between the parties likely to account for the pattern. If the minority stake accompanied joint research in the shared classes, patenting should shift toward joint filings of the acquirer and the target. Joint applications are instead rare and do not increase. In fewer than five percent of the overlapping minority stake acquisitions in our sample do the acquirer and target ever file a patent together, and in only 1.2 percent do they do so in the five years after the acquisition; joint filings are flat to declining around the deal.

6.4 The Importance of the Shared Technology Space

Our definition of overlap is deliberately coarse. A single shared 4-digit IPC class is enough to classify an acquisition as overlapping. Yet the shared classes may be the core of a firm's research or a marginal sideline, and the two parties need not be equally exposed. The externality at the heart of Section 3 requires both. The acquirer's innovation can only depress the target's profit to the extent that the target actually earns in the contested classes, and the target can only respond to the extent that its own returns move with what the acquirer does there.

We therefore measure, for each party, the share of its pre-acquisition patents that falls into the shared classes, and take the smaller of the two. This "shared-space importance" is high only when the contested classes are central for *both* firms; it is low whenever they are peripheral for either one. We then split the sample at the median of this measure across deals (16%), separating deals in which the shared classes account for an above-median share of both firms' pre-acquisition patenting from the rest.[30]

The outcome here is again total patenting at the firm level, as in the baseline specification of Table 1, and not one of the two portfolio components of the previous figures; what varies across the two groups is the kind of deal, not the set of classes we count.

Figure 4 shows that the two sides respond to different conditions. For targets, the effect is concentrated in the deals where the shared space matters to both parties. The year-five effect is -8.7% in that group ($n = 639$), while the remaining deals show no measurable response at any horizon ($n = 921$).[31] For acquirers, by contrast, the split makes no difference. Both groups decline significantly, by 10–13% by year five, bracketing the full-sample estimate of Figure 1.

[30]We apply the same cutoff to acquirers and targets. Side-specific medians differ considerably (10% for targets, 25% for acquirers) and would classify the same deal as material for one party but immaterial for the other, mechanically generating asymmetries between the two sides.

[31]The two groups comprise fewer deals than the matched sample because the measure is undefined whenever the counterpart cannot be identified in the deal data or holds no pre-acquisition patents; such deals cannot be classified and are excluded from the split.

The conditional response of targets is also hard to square with a reading of minority stakes as primarily vehicles for technology access. Access motives may well be present, but they would give the target no reason to cut back on its own research—least of all in precisely those deals in which the shared space matters most to both sides.

Targets and acquirers thus respond under different conditions. The target moves only where the shared space matters to both sides, which is where a reaction to the counterpart's behavior should show up. Acquirers cut back in both groups, and that speaks against duplication as the driving force. In deals whose shared classes are marginal to one of the two research programs there is little duplicated work to eliminate, yet acquirers retreat by just as much.

The median cutoff is not a knife edge. In Figure A.9 in the Appendix we vary the cutoff from 10% to 50% of both portfolios. The pattern is the same throughout. The year-five effect for the above-cutoff target group lies between -7% and -9% at every cutoff, the below-cutoff target group stays between -2% and -4% and is statistically indistinguishable from zero except at the 50% cutoff, where that group comprises most of the sample, and acquirers decline significantly in both groups at every cutoff. What separates the two groups of targets is thus not how much technology the two firms share, but whether the technology they share is material to both of them.

This also puts our headline estimates in perspective. Because the binary overlap definition counts any shared class, the target sample mixes deals with no measurable response together with deals with a year-five cumulative gap of about nine percent. We retain the binary definition as our main specification because it is transparent and can be applied to every deal, but the average target effect we report understates the response of the firms whose research programs genuinely collide.

Product-market overlap sharpens this picture. Figure A.19 in the Appendix splits the technologically overlapping acquisitions by whether the direct deal parties also share a four-digit NACE industry code. For targets, the decline is concentrated among acquisitions that overlap in both technology and product markets. Five years after the acquisition, the estimated effect is -8.5% in that group, compared with -3.3% in the absence of product-market overlap. For acquirers, the estimates are negative and similar in magnitude in both cells. Product-market overlap thus appears to strengthen the target response, while it is not a necessary condition for the acquirer response. The pre-acquisition coefficients are small in all four groups.[32]

[32]The pattern is not specific to NACE. Defining industry overlap through four-digit SIC or NAICS codes, or combinations of SIC, NACE, and NAICS, yields five-year target estimates between -8.5% and -10.1%

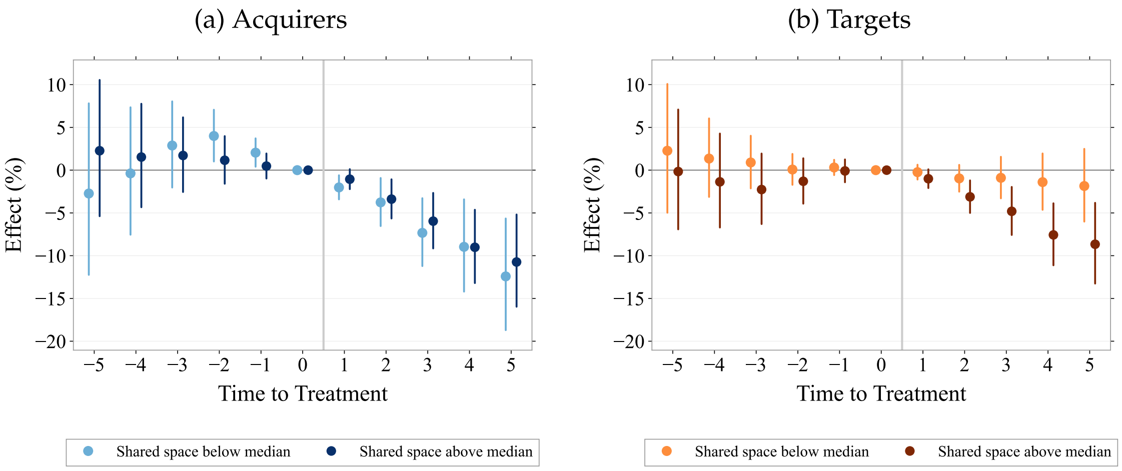

Figure 4: Event Studies by the Importance of the Shared Technology Space for Both Parties

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms, estimated separately for two groups of deals. For each deal we compute the share of each party's pre-acquisition patents that falls into the classes the two firms share, and take the smaller of the two shares. Dark squares: deals for which this share is above its sample median of 16%, i.e. the shared classes account for an above-median share of both firms' pre-acquisition patenting. Light circles: all remaining deals. Control firms enter each group with the weight of their matched treated firms in that group. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. The estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each series shows 95% confidence intervals based on standard errors clustered at the firm level.

6.5 Citation-Weighted Innovative Output

Patent counts weight every patent equally. If the decline documented above reflected a retreat from marginal, low-value filings, weighting patents by their subsequent impact should reveal a smaller effect. We therefore re-estimate the event studies with the log of the cumulative number of forward citations as the dependent variable.

Figure 5 shows that citation-weighted output falls at least as strongly as patent counts. By year five, citation-weighted output is 14.1% lower for acquirers and 5.3% lower for targets. The reduction in patenting is thus not confined to patents that attract little subsequent attention—if anything, the quality-weighted margin falls more steeply. For targets, the pre-acquisition coefficients are flat throughout. For acquirers, the coefficient in the year before the acquisition is positive; the sensitivity analysis in Section 6.6, which explicitly allows for differential pre-trends, covers the citation results as well.

with product-market overlap and between -2.7% and -3.3% without, and acquirer estimates between -9.0% and -9.4% and between -11.9% and -12.8%, respectively.

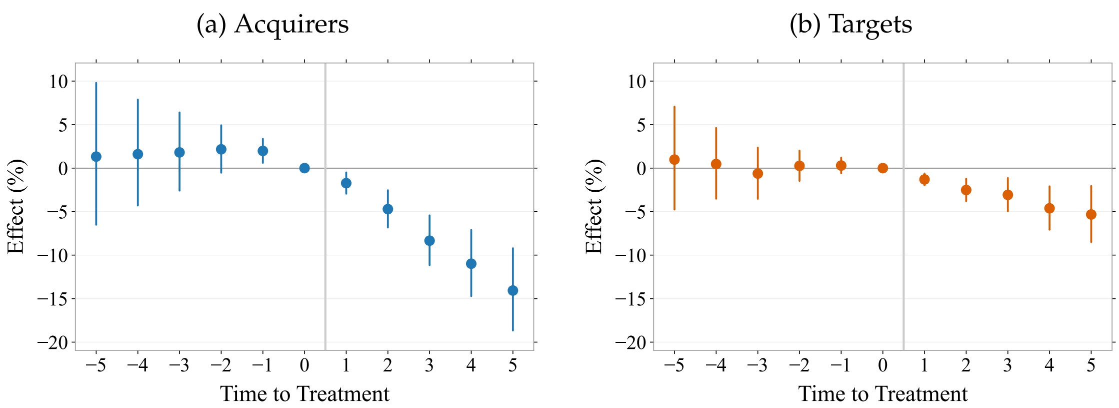

Figure 5: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Citation-Weighted Patent Output

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. The dependent variable is the log of the cumulative number of forward citations to the firm's granted patents, per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

6.6 Assessing the Credibility of the Main Estimates

Flat estimated pre-trends do not by themselves rule out violations of parallel trends, and small pre-existing differences could in principle bias the post-treatment estimates in either direction (Roth et al., 2023; Rambachan and Roth, 2023).

We address this concern in two ways and then subject the estimates to a series of further checks. First, following Fenizia and Saggio (2024), we fit a linear trend to the pre-treatment coefficients and extrapolate it into the post-treatment period (left panels of Figure 6). For acquirers the estimated slope is essentially zero, and rotating the event-study coefficients around the extrapolated trend (right panels) leaves the estimates virtually unchanged. For targets the fitted slope is -0.0021 log points per year, so the rotation shifts the post-treatment path toward zero; the rotated estimates remain negative and significant from the second post-treatment year onward. The rotation is a descriptive device that does not account for the estimation uncertainty of the fitted trend. The Rambachan–Roth analysis below provides the corresponding formal inference. At $M = 0$ its robust confidence intervals are equivalent to removing an estimated linear trend while fully accounting for that uncertainty, and for $M > 0$ they additionally allow for non-linear deviations.

Second, we implement the "honest" difference-in-differences approach of Rambachan and Roth (2023), which bounds the permissible change in the slope of the differential trend between treated and control firms:

$$\Delta\text{Slope Differential} = \{\theta : |(\theta_{k+1} - \theta_k) - (\theta_k - \theta_{k-1})| \leq M\} \,.$$

The parameter M controls how much the slope of the differential trend is allowed to bend in each post-treatment period: the effect remains identified as long as the counterfactual trend deviates from a straight line by at most M percentage points per period, with $M = 0$ permitting an arbitrary *linear* differential trend. We report the fifth post-treatment coefficient, where cumulative deviations from linearity are largest and the test is most demanding; the earlier coefficients face smaller cumulative deviations and are correspondingly less sensitive.

Allowing for arbitrary linear violations of parallel trends ($M = 0$), the patent effects at $t + 5$ remain clearly bounded away from zero for both acquirers (robust confidence interval $[-12.9\%, -4.0\%]$) and targets ($[-7.0\%, -1.9\%]$). Both effects thus survive even a full continuation of the fitted pre-acquisition trends. Beyond that, the counterfactual slope may steepen by a further 0.19 (acquirers) and 0.14 (targets) percentage points in every post-acquisition period before the year-five intervals include zero. For targets, each year of permitted bending amounts to about two thirds of the entire fitted pre-acquisition slope, so that by year five the counterfactual slope may steepen to more than three times the fitted slope. For acquirers, whose fitted pre-trend is essentially flat, the permitted deviations are large relative to anything estimated in the pre-period. The robust confidence intervals for a grid of M values are reported in Figure A.8 in the Appendix. The same sensitivity analysis applied to citation-weighted output yields the same conclusion.[33]

[33] At $M = 0$ the robust confidence intervals at $t + 5$ for citation-weighted output are $[-14.5\%, -4.3\%]$ for acquirers and $[-8.0\%, -1.2\%]$ for targets.

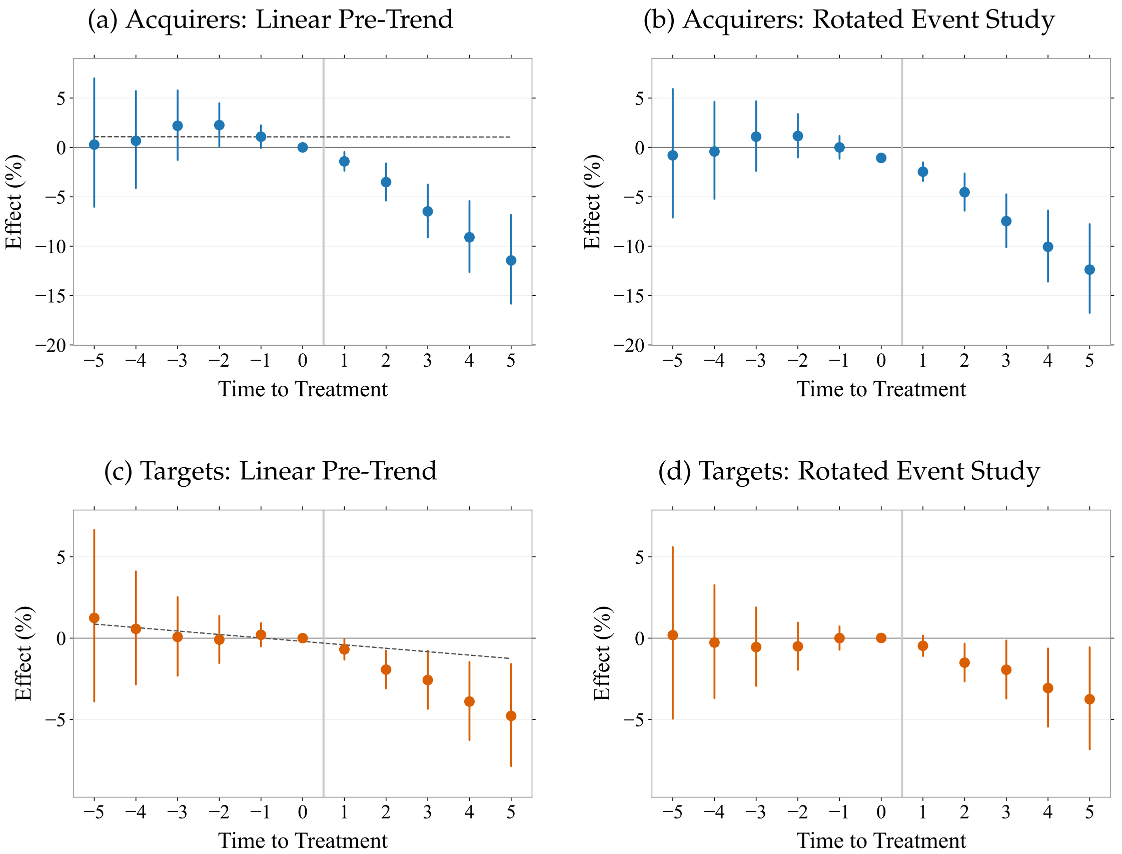

Figure 6: Pre-Trend Extrapolation and Rotated Event Studies (Patents)

The left panels show the patent event studies together with a linear trend (dashed line) fitted by OLS to the pre-treatment coefficients (five years before the minority stake acquisition, including the reference period) and extrapolated into the post-treatment period. The right panels rotate the event-study coefficients around this extrapolated trend, re-anchored at the reference period. Confidence intervals (95%, standard errors clustered at the firm level) in the right panels are those of the original event-study coefficients, shifted with the rotation; they do not incorporate the estimation uncertainty of the fitted trend, which is addressed formally by the Rambachan–Roth sensitivity analysis in the text.

Third, we verify that the results are not an artifact of pointwise inference. The confidence intervals in Figure 1 cover each coefficient individually with 95% probability, but not the path of all ten coefficients jointly. Following the recommendation of Baker et al. (2025), we compute sup-t simultaneous confidence bands (Montiel Olea and Plagborg-Møller, 2019), which cover the entire coefficient path with 95% probability. The resulting critical values, obtained from 200,000 draws of the estimated correlation matrix of the coefficients, are 2.61 (acquirers) and 2.56 (targets) instead of the pointwise 1.96. Figure 7 shows the corresponding widening of the bands. The conclusions are unchanged. No pre-treatment coefficient reaches the simultaneous threshold—the pre-treatment path is jointly indistinguishable from zero—while the post-treatment decline remains significant under the simultaneous bands from the first post-treatment year onward for acquirers and from the second year onward for targets. In particular, the simultaneous bands exclude zero for the year-five effects of both groups ($|t| = 4.7$ and 2.9).

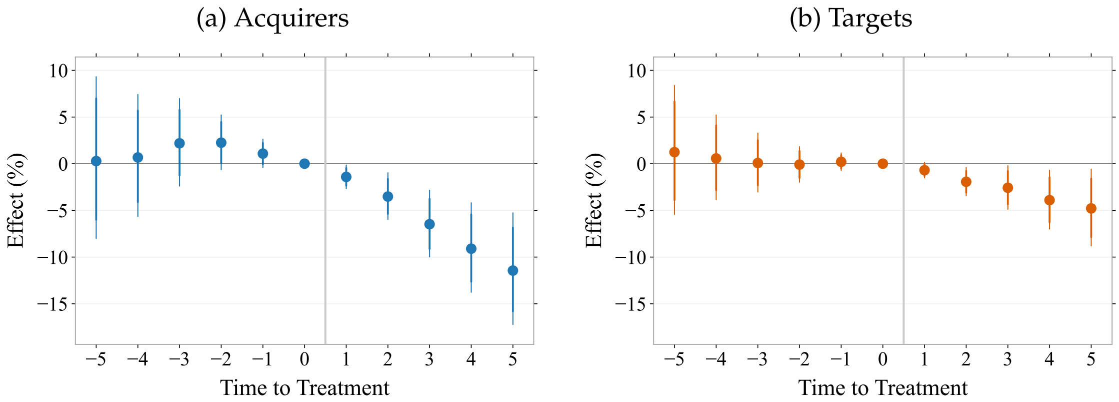

Figure 7: Sup-t Simultaneous Confidence Bands (Patents)

The figure repeats the patent event studies of Figure 1. The inner (light) bars are the pointwise 95% confidence intervals; the green extensions widen them to sup-t simultaneous 95% confidence bands (Montiel Olea and Plagborg-Møller, 2019), which cover the entire path of event-study coefficients jointly with 95% probability. Critical values are computed from 200,000 draws from the estimated correlation matrix of the coefficients. Standard errors are clustered at the firm level.

Excluding Japan and Korea. As discussed in Section 4, minority shareholdings play a special institutional role in Japan and Korea. Cross-held minority stakes are a defining feature of the *keiretsu* business groups and "stable shareholder" relationships in Japan and of the *chaebol* conglomerates in Korea. Minority stakes in these environments may thus reflect group affiliation and relational governance rather than a strategic acquisition of a stake in a rival, and the two countries together account for roughly half of the overlapping acquisitions in our sample. To verify that the results are not driven by these institutionally distinct environments, we re-estimate the model excluding all treated firms (and their matched controls) from Japan, from Korea, and from both countries jointly. Figure 8 reports the average effects over the five-year window (top panels) and the year-five effects (bottom panels) on patents across the four samples. The results are not driven by Japan or Korea. Every exclusion leaves both the average and the year-five effects statistically significant for both groups. If anything, the effects strengthen when Japan is excluded. This pattern is consistent with the institutional argument above. To the extent that cross-held stakes within business groups are less likely to reflect a strategic acquisition of a stake in a rival, including them dilutes the estimated effect.

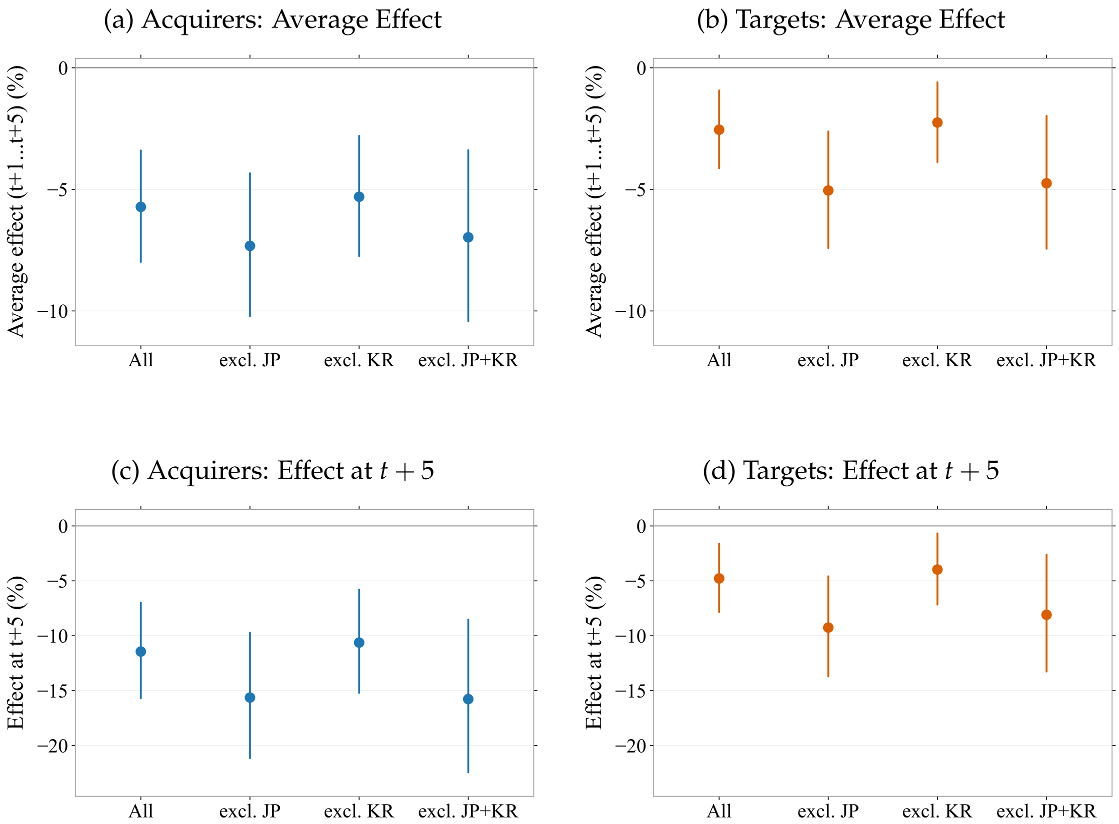

Figure 8: Effects on Patents When Excluding Japan and Korea

The figure shows the estimated average effect on cumulative patents over the five post-acquisition years (top panels) and the effect five years after the acquisition (bottom panels) in four samples: the full sample, excluding Japan, excluding Korea, and excluding both. Each exclusion drops all treated firms headquartered in the respective country together with their matched control firms. Points are the estimates, bars the 95% confidence intervals based on standard errors clustered at the firm level.

Spillovers to Control Firms. One concern is that the comparison itself may be contaminated by the deal. The matched control firms patent in the same 4-digit IPC classes as the treated firms, and if the acquisition changes competitive conditions in these fields, such technologically close controls may themselves respond, so that the comparison would partly reflect this response rather than the behavior of unaffected firms. We address this concern by re-matching treated firms to controls that are active in the same 3-digit IPC classes but share no 4-digit class with the treated firm. These controls remain technologically comparable while standing outside the directly affected fields. The acquirer effect is essentially unchanged under this matching, with a year-five effect on patents of -11.6% (Figure A.13 in the Appendix), indicating that the baseline estimates do not hinge on indirectly affected controls. The target effect is smaller (-2.9%) and less precisely estimated, in line with the substantially reduced pool of admissible controls.

Escalation to Majority Control. The baseline sample excludes acquisitions in which the acquirer obtains majority control of the target within five years, so that the estimates are not confounded by outright takeovers. A second concern is that this exclusion conditions the sample on a post-acquisition event. We therefore re-run the analysis retaining these deals and instead censoring the affected firms from the year the stake turns into a majority position, so that their minority-stake years remain in the panel. The matched sample grows to 2,128 targets and 1,004 acquirers, and the year-five effects on patents are -10.3% for acquirers and -3.5% for targets, close to the baseline estimates (Figure A.14 in the Appendix).

Overlap Defined by Industry Codes. Our main specification defines rivalry by proximity in the technology space. As an alternative, we classify an acquisition as overlapping if the acquirer and target share at least one 4-digit NACE industry code, and match on industry codes rather than IPC classes. The estimates remain negative and significant, reaching -4.8% for acquirers and -5.9% for targets by year five (Figure A.16 in the Appendix). For targets the magnitude is comparable to the technology-based estimate, for acquirers it is roughly 40 percent of it. This is consistent with the mechanism operating through technological rivalry. The externality of Section 3 runs through innovation in overlapping technologies, so technological overlap should be the sharper screen for innovation responses, with product-market overlap a coarser proxy.

In sum, none of these checks moves the estimates. They are driven neither by the institutionally distinct Japanese and Korean environments nor by indirectly affected control firms, and they survive both the reinclusion of acquisitions that escalate to majority control and the industry-code definition of rivalry.

7 Conclusion

Firms may compete less aggressively through innovation once one firm owns part of another. When the acquirer and target operate in overlapping technology fields, the acquirer bears part of the loss that its own innovation imposes on the target, weakening its incentive to innovate against it. The target may then respond to the acquirer's pullback by reducing its own innovation. Consistent with this mechanism, both acquirers and targets patent less after minority stake acquisitions between such firms. By the fifth post-acquisition year, cumulative patent output is roughly 11% lower for acquirers and 5% lower for targets than for their matched controls. Citation-weighted output follows a similar path. We find no comparable decline after minority stake acquisitions between firms whose technologies do not overlap.

The decline is closely related to where the two firms innovate. Both firms reduce patenting in the technology classes they share. Outside the shared space, targets' patenting rises relative to matched controls, whereas acquirers reduce patenting there as well. Minority ownership therefore affects not only how much firms patent, but also where they patent.

Acquirers and targets respond under different conditions. Targets cut back mainly when the shared technology space matters to both firms, as one would expect if the target reacts to a change in the acquirer's behavior rather than to the deal itself. Acquirers reduce patenting even when the shared classes are marginal to one party's research program, leaving little duplicated R&D to eliminate. Taken together, these patterns are more consistent with weakened competitive incentives than with financial constraints, managerial distraction, or the elimination of duplicative R&D.

Minority stake acquisitions are common, but their effects on innovation have received little empirical attention. The findings are relevant for competition policy because partial acquisitions often receive less scrutiny than full mergers, even though all transactions in our sample remain below majority ownership and most stakes are below 25%. At the same time, the results do not imply that every minority acquisition warrants closer review. We find no comparable innovation effect when the firms' technologies do not overlap, and reviewing every minority investment would impose substantial costs on agencies and firms. Technological overlap can therefore help identify the transactions in which innovation incentives are most likely to change, while the importance of the shared fields and product-market overlap between the parties help identify where target responses are strongest. More broadly, our results show that minority ownership can reshape both the intensity and the direction of innovation even when the ownership stake remains far below a majority position.

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A Appendix

A.1 Sample Construction and Representativeness

Table A1 details how we constructed the datasets for acquirers and targets that we use in the paper and the number of firms that remained after each step. To ensure that our selected group of firms constitutes a representative sample, we compare our group of treated firms with the entire set of companies involved in minority stake acquisitions as reported in the Zephyr dataset. The distribution of our final sample across years is similar to other firms involved in minority stake acquisitions (see Figure A.1). The later years (2018–2019) are under-represented in our sample because we require that treated firms have at least three years of consecutive patent data. As shown in Figure A.2, the distribution of our final sample across countries is roughly similar to other firms involved in minority stake acquisitions. China is under-represented in our sample, while Japan and South Korea are over-represented. The distribution of our final sample across industries is also similar to other firms involved in minority stake acquisitions (see Figure A.3). Similar to other firms, our sample of acquirers and targets predominantly operate in the manufacturing sector. As expected, our sample is under-represented in terms of Finance, Insurance, and Management companies, since we excluded non-corporate firms and investment offices from the group of treated firms.

A.2 Figures

(a) Sample Overlapping Minority Acquisitions

(b) All Minority Acquisitions

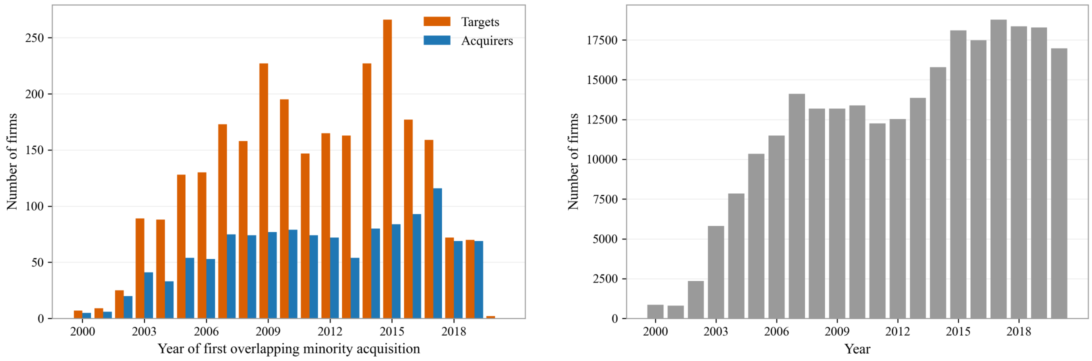


Figure A.1: Distribution of Minority Stake Acquisitions Over Time

Panel (a) shows the distribution of the number of targets and acquirers in our sample involved in overlapping minority stake acquisitions by year, over the sample period. Panel (b) shows the distribution of all firms involved in any minority stake acquisition by year for the period from 2000 to 2020, as per the Zephyr database.

(a) Sample Overlapping Minority Acquisitions

(b) All Minority Acquisitions

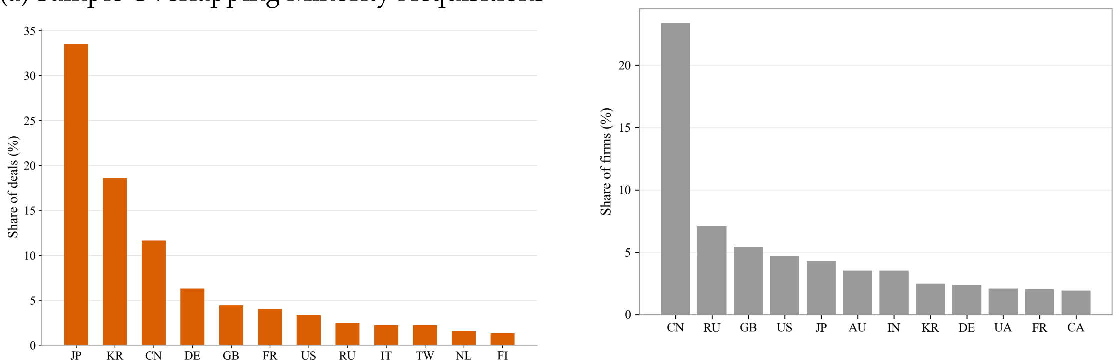


Figure A.2: Distribution of Minority Stake Acquisitions Across Countries

Panel (a) shows the distribution of the overlapping minority stake deals in our sample by the country of the treated firm, pooling targets and acquirers. Panel (b) shows the distribution of all firms involved in any minority stake acquisition by country, as per the Zephyr database.

(a) Sample Overlapping Minority Acquisitions (b) All Minority Acquisitions

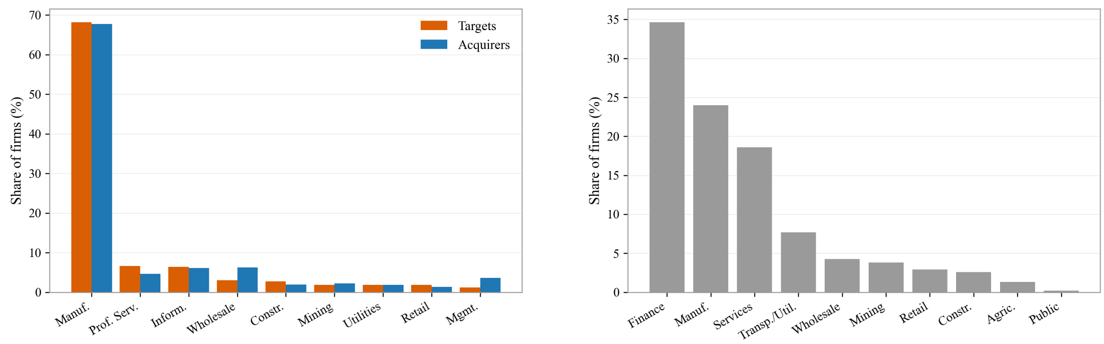


Figure A.3: Distribution of Minority Stake Acquisitions Across Industries

Panel (a) shows the distribution of the number of targets and acquirers in our sample involved in overlapping minority stake acquisitions by sector (based on 2-digit NAICS codes). Panel (b) shows the distribution of all firms involved in any minority stake acquisition by sector (based on the primary US SIC code reported in the Zephyr database).

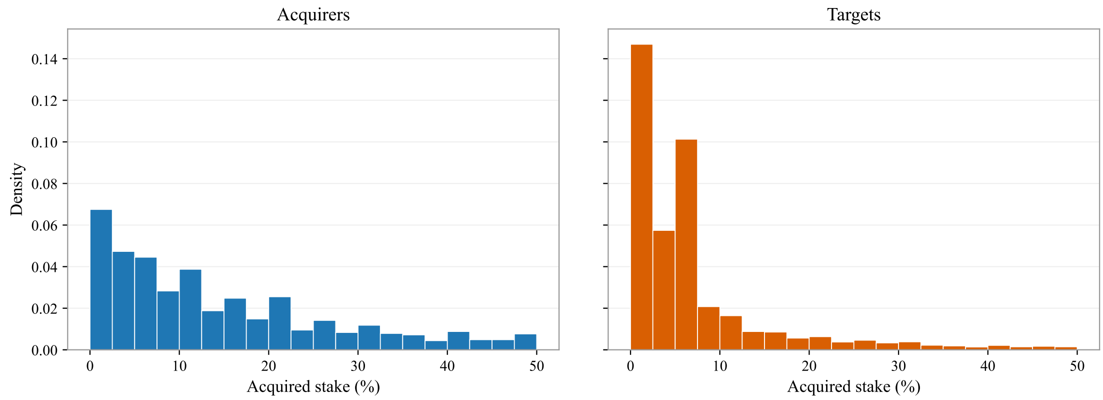


Figure A.4: Distribution of Overlapping Minority Stakes in Our Sample

This figure shows the distribution of the acquired overlapping minority stakes for acquirers and targets included in our sample, measured in the year of the acquisition as used in the treatment definition. Deal-years with missing stake information or stakes above 50% are excluded.

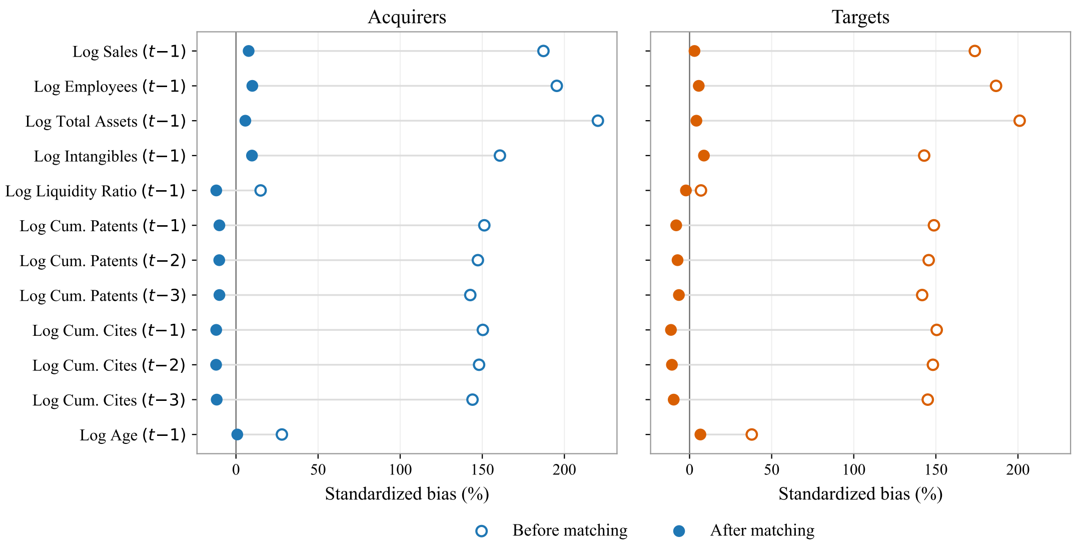
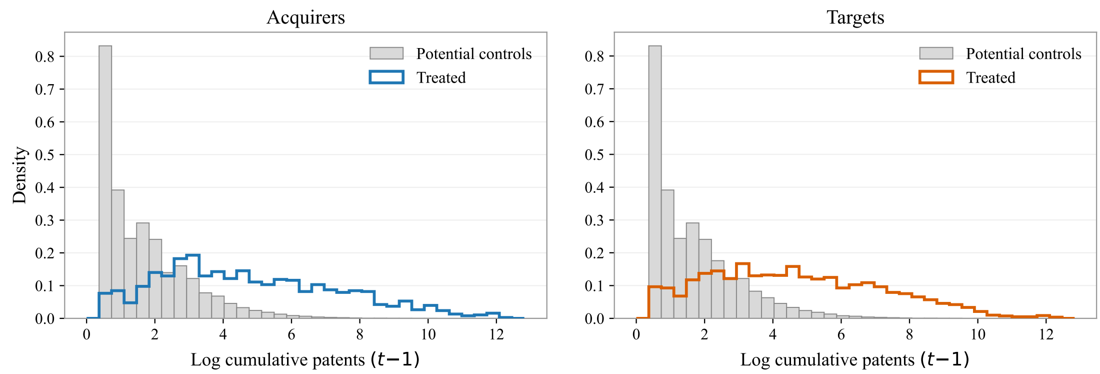


Figure A.5: Standardized Biases, Before and After Matching

This figure plots the standardized biases for the covariates used in the propensity score estimation, before and after matching, separately for targets and acquirers. For each regressor, the standardized bias is calculated as the difference of sample means in the treated and control samples as a percentage of the square root of the average of sample variances in both groups (Caliendo and Kopeinig, 2008). Before matching, biases are computed using all firm-year observations from the pool of potential controls alongside the $t-1$ observation for treated firms. After matching, biases are computed using the $t-1$ observation for matched treated and control firms only. The matched firms are then followed as a panel for the event-study estimation.

(a) Before Matching



(b) After Matching

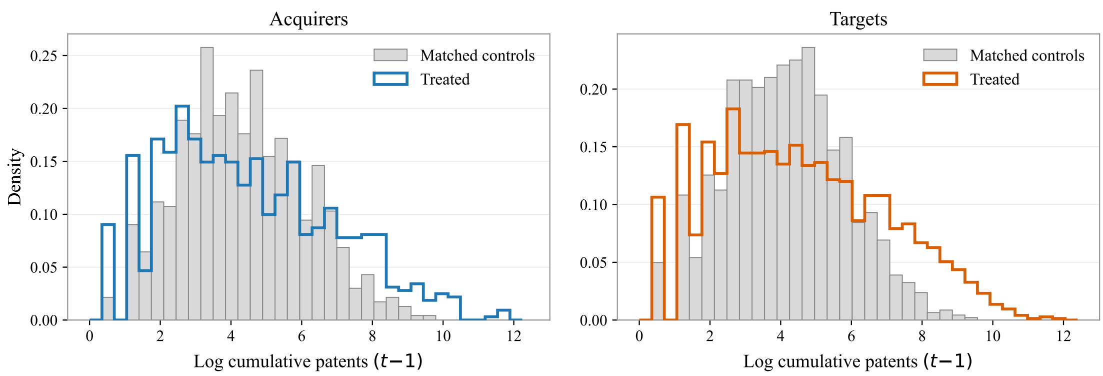


Figure A.6: Distribution of Patents for Treated and Control Firms, Before and After Matching

The top panels plot the distribution of the log of the cumulative number of patents for treated firms (the year before the acquisition) and all potential control firms. The bottom panels plot the distribution of the log of the cumulative number of patents for the matched treated and control firms (the year before the acquisition).

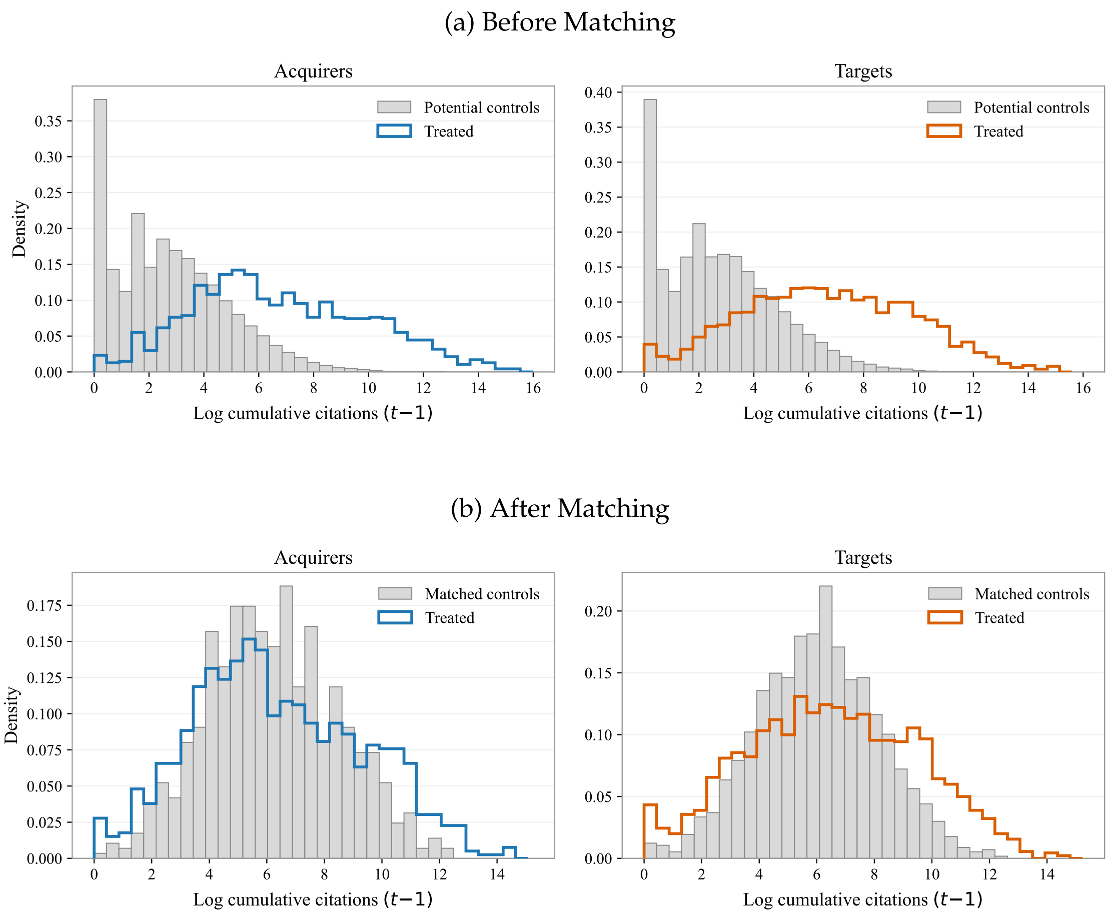


Figure A.7: Distribution of Patent Citations for Treated and Control Firms, Before and After Matching

This figure compares the distribution of patent citations for treated and control firms, before and after the matching. The top panels plot the distribution of the log of the cumulative number of patent citations for treated firms (the year before the acquisition) and all potential control firms. The bottom panels plot the distribution of the log of the cumulative number of patent citations for the matched treated and control firms (the year before the acquisition).

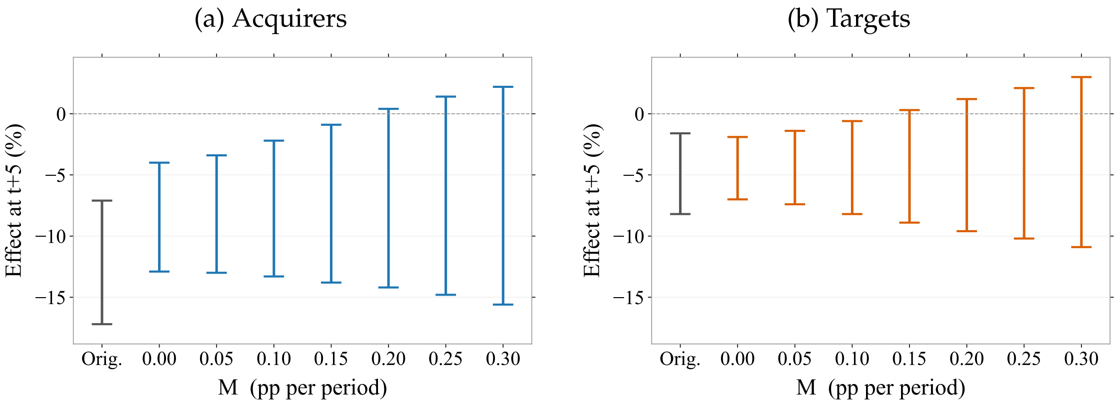


Figure A.8: Robust Confidence Intervals for the Year-Five Patent Effect Under Bounded Violations of Parallel Trends

The figure reports Rambachan and Roth (2023) robust 95% confidence intervals for the effect on cumulative patents five years after the overlapping minority stake acquisition, as a function of the sensitivity parameter M (horizontal axis, in percentage points per period). M bounds the period-to-period change in the slope of the differential trend between treated and control firms: $M = 0$ allows an arbitrary linear differential trend, and larger values additionally allow non-linear deviations from linearity. "Orig." denotes the original event-study confidence interval. The robust intervals are fixed-length confidence intervals computed from the event-study coefficients with five pre-treatment (placebo) coefficients; standard errors are clustered at the firm level.

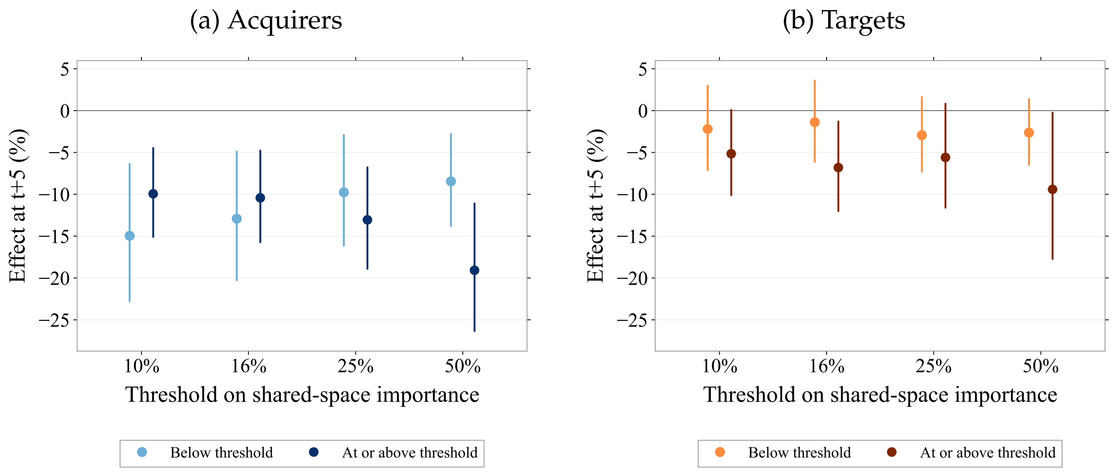


Figure A.9: Year-Five Effects by the Threshold on Shared-Space Importance

The figure reports the estimated effect on cumulative patents five years after the acquisition, separately for deals above and below a threshold on the importance of the shared technology space, for four different thresholds. For each deal we compute the share of each party's pre-acquisition patents that falls into the classes the two firms share and take the smaller of the two shares; the threshold is applied to that measure and is identical for acquirers and targets. Each treated firm's matched control is assigned to the same group. Group sizes for acquirers (below/above) are 253/666, 346/573, 460/459 and 655/264; for targets 813/747, 921/639, 1057/503 and 1269/291. The mean of the measure in the group above the threshold rises from 0.46 to 0.75 across the four cutoffs, and in the group below it from 0.03 to 0.14. Estimates use the estimator presented in de Chaisemartin and D'Haultfoeuille (2026) with firm and year fixed effects; bars are 95% confidence intervals based on standard errors clustered at the firm level.

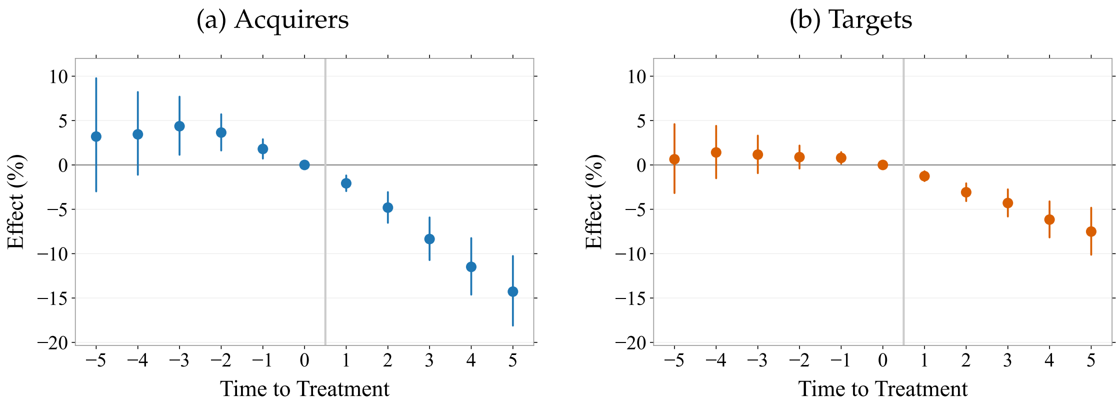


Figure A.10: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Unweighted Estimation

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, the estimation does not weight control firms by the number of treated firms they serve as a match for; the matched sample is identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

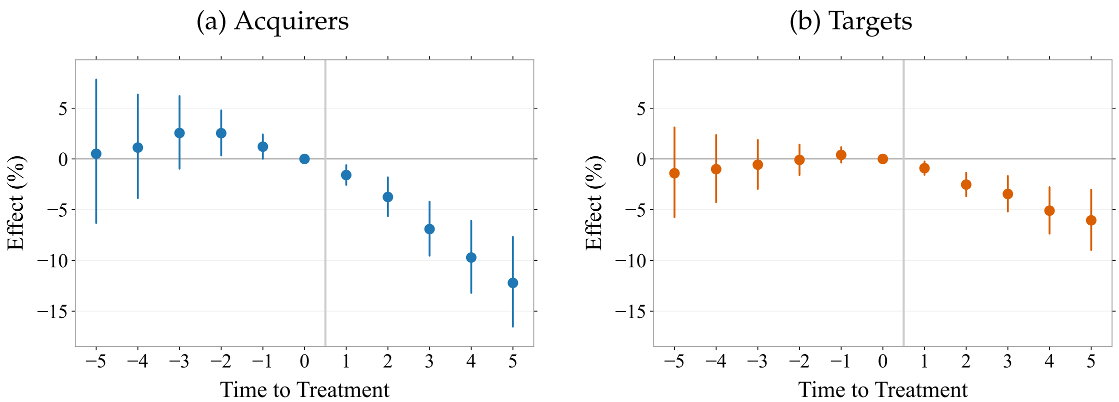


Figure A.11: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Matching Without Replacement

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, the matching is performed without replacement, so that each control firm serves as the match for at most one treated firm; all other elements of the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

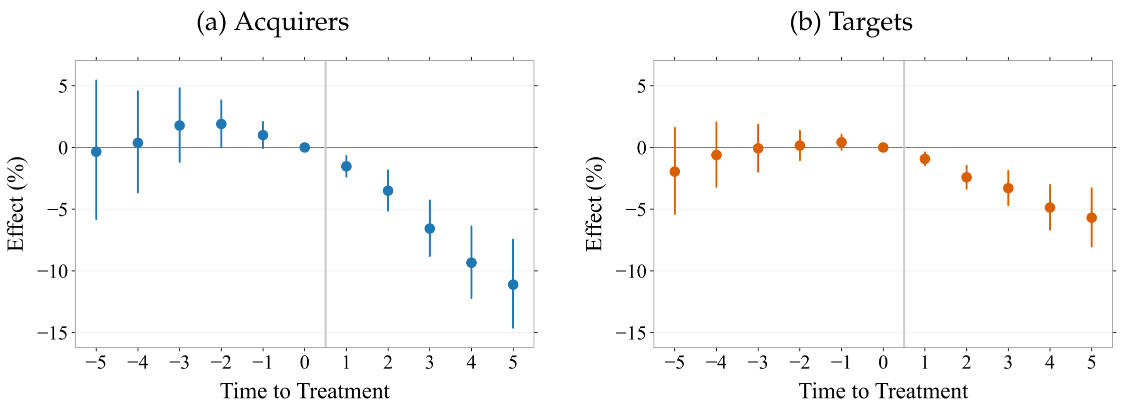


Figure A.12: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Matching on Three Nearest Neighbors

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, each treated firm is matched to its three nearest neighbors in terms of the propensity score instead of one; all other elements of the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

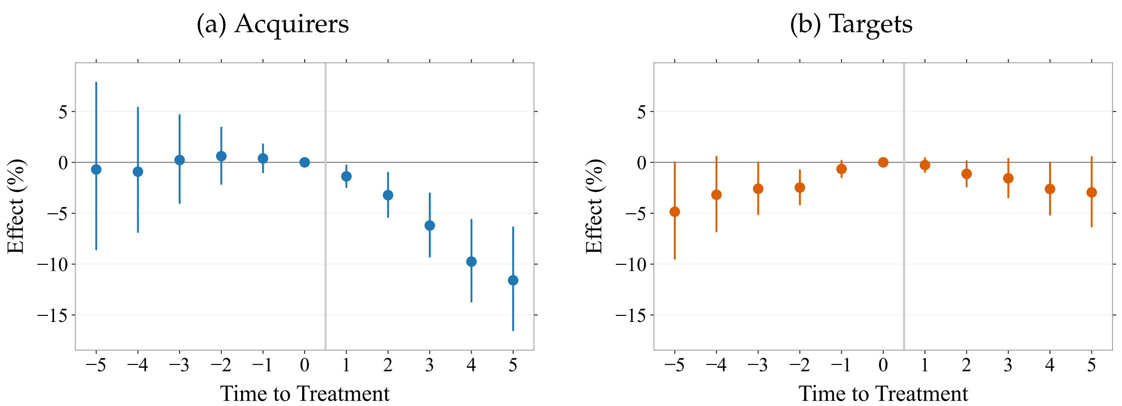


Figure A.13: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Matching on 3-digit IPC Overlap Excluding 4-digit Overlap

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, the control group is matched on 3-digit IPC overlap while excluding firms that share any 4-digit IPC class with the treated firm. This ensures that control firms operate in the same broad technology area but their technologies do not overlap those of the treated pair. All other elements of the matching procedure (propensity score covariates including the lagged innovation levels, calipers, and common support) are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

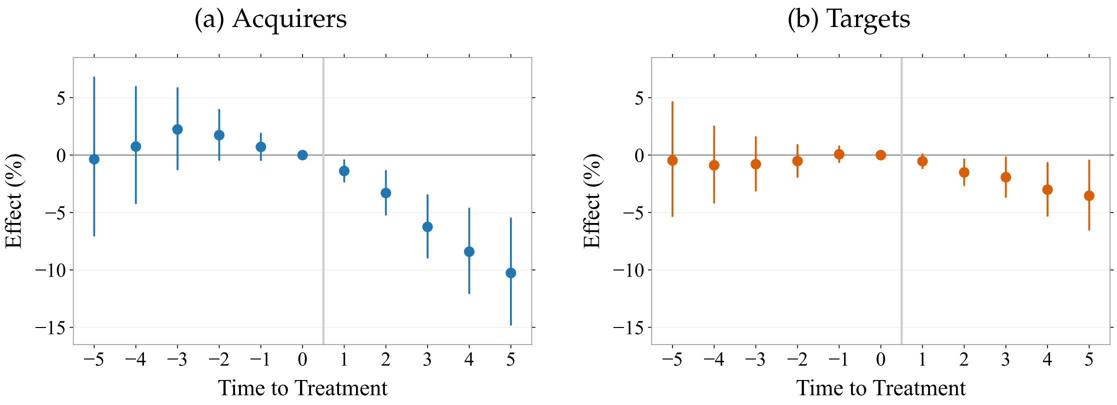


Figure A.14: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Retaining Deals that Escalate to Majority Control

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, deals in which the acquirer obtains majority control of the target within five years of the minority acquisition are retained; the affected firms are instead censored from the year the stake turns into a majority position. All other elements of the sample construction and the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

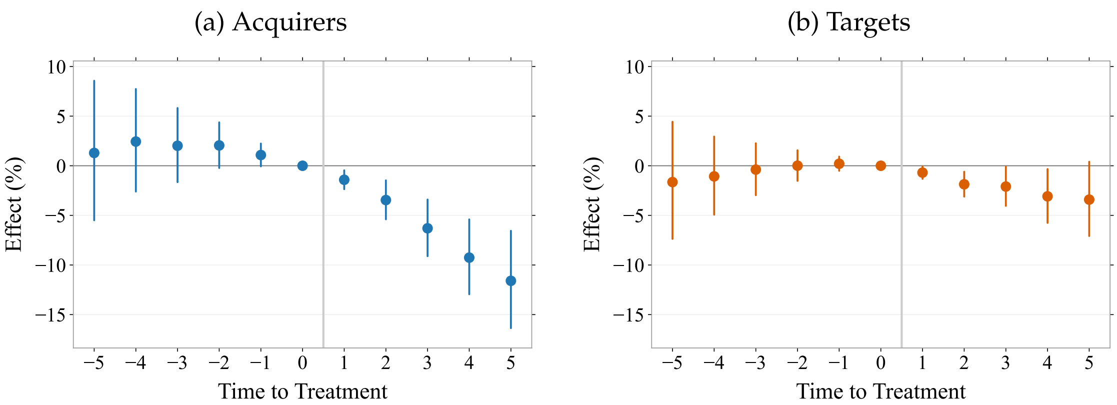


Figure A.15: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Censoring at the Second Overlapping Acquisition

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, treated firms are censored from the year of their second overlapping minority stake acquisition onward, so that the estimates reflect exposure to the first acquisition only; all other elements of the sample construction and the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

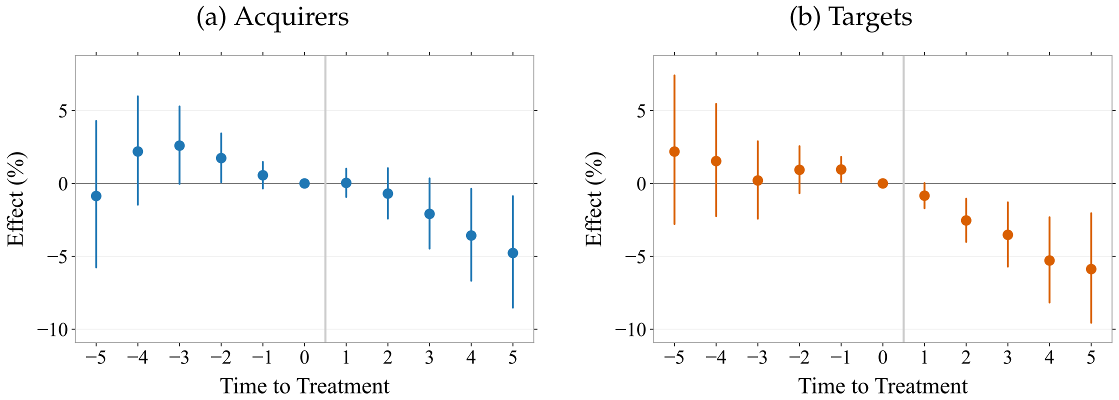


Figure A.16: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output—Overlap Defined by Industry Codes

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. Overlapping deals are defined by an overlap in the 4-digit NACE industry codes of the target and its acquirer/parent; the matching procedure likewise requires industry-code overlap instead of IPC overlap. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

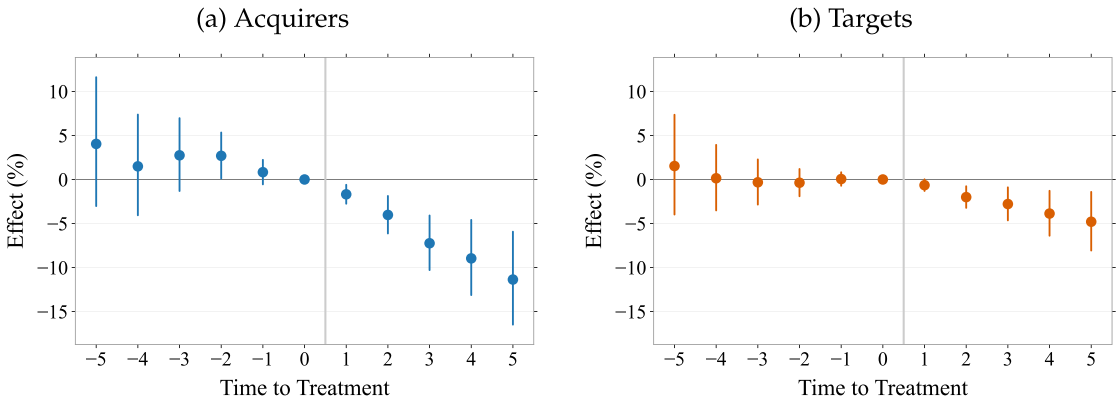


Figure A.17: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Stakes below 20 Percent

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure 1, the treated sample is restricted to firms whose stake acquired in the first overlapping acquisition is below 20 percent, covering 89 percent of treated targets and 67 percent of treated acquirers; control firms follow their matched treated firm. All other elements of the sample construction and the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

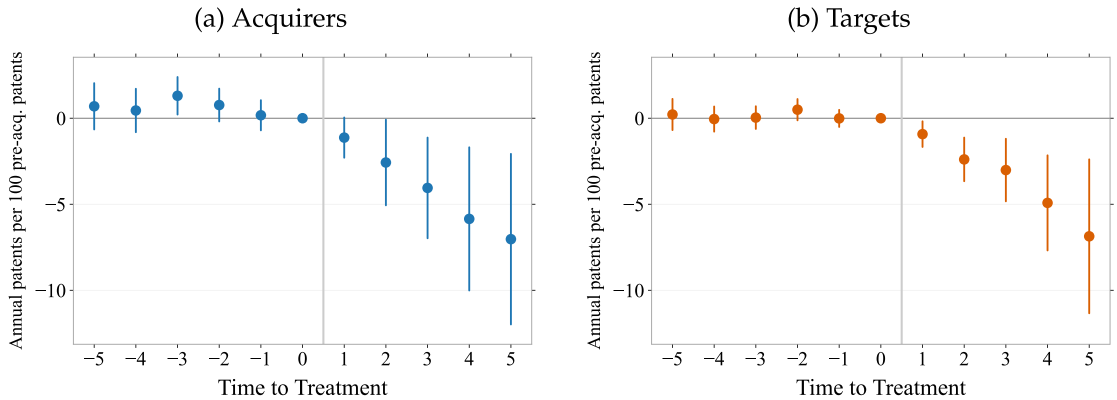


Figure A.18: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Annual Patent Production

The figure presents coefficient estimates for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. The dependent variable is the number of granted patents (dated by application year) filed by the firm in year t, divided by one plus the firm's cumulative number of granted patents in the last pre-acquisition year, expressed per 100 pre-acquisition patents. Estimates use the same matched sample and the estimator presented in de Chaisemartin and D'Haultfoeuille (2026), with each control firm aligned to the acquisition cohort of its matched treated firm. Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

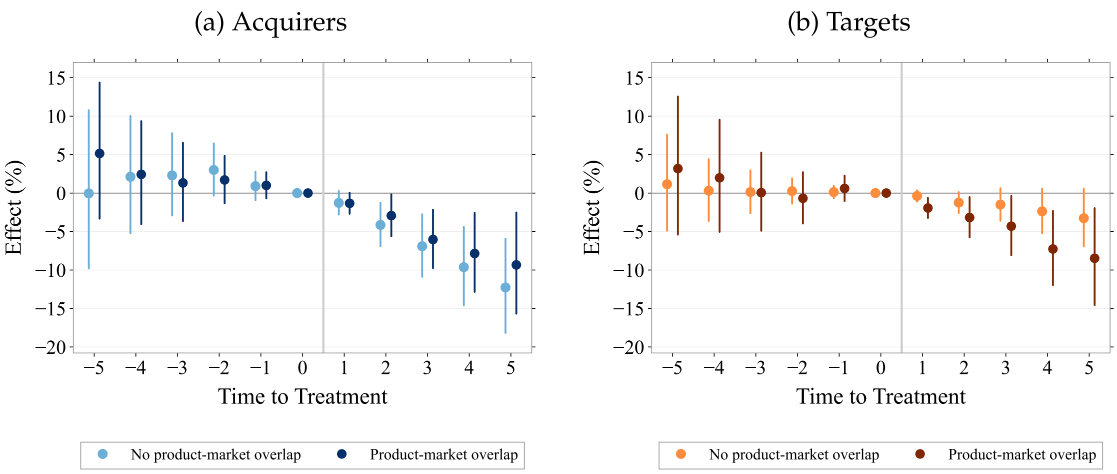


Figure A.19: Event Studies by Product-Market Overlap of the Deal Parties

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms, estimated separately for two groups of deals. Product-market overlap means that the direct deal parties share at least one native four-digit NACE code; both parties must have comparable NACE information, and deals with unknown status are excluded rather than assigned to the no-overlap group (246 targets and 254 acquirers). Control firms enter each group with the weight of their matched treated firms in that group. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. The estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026), weighted by match weights. Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each series shows 95% confidence intervals based on standard errors clustered at the firm level.

A.3 Tables

Table A1: Dataset Construction

	Action	Remaining Companies: Targets	Acquirers
Step 1	Firms involved in overlapping minority stake acquisitions	4,561	2,898
Step 2	Eliminate deals in which the acquirer obtains majority control of the target within 5 years of the minority acquisition	4,345	2,734
Step 3	Eliminate firms without Orbis financial data	2,783	1,311
Step 4	Eliminating non-corporate firms (e.g., financial company, mutual and pension fund)	2,730	1,261
Step 5	Eliminate firms without 3 years of consecutive patent data	2,682	1,239

This table details how we constructed the datasets for targets and acquirers that we use in the paper and the number of firms that remained after each step.

Table A2: Propensity Score Estimation (Probit Model)

	Targets (1)	Acquirers (2)
Log Sales ($t-1$)	-0.186 (0.014)***	-0.160 (0.019)***
Log # Employees ($t-1$)	0.226 (0.013)***	0.130 (0.018)***
Log Total Assets ($t-1$)	0.266 (0.014)***	0.341 (0.019)***
Log Intangibles ($t-1$)	0.044 (0.003)***	0.053 (0.005)***
Log Liquidity Ratio ($t-1$)	0.141 (0.020)***	0.249 (0.026)***
Log Cum. Patents ($t-1$)	0.139 (0.102)	0.227 (0.144)
Log Cum. Patents ($t-2$)	-0.044 (0.150)	-0.435 (0.207)**
Log Cum. Patents ($t-3$)	0.009 (0.089)	0.329 (0.121)***
Log Cum. Patent Citations ($t-1$)	0.111 (0.065)*	-0.051 (0.110)
Log Cum. Patent Citations ($t-2$)	0.011 (0.085)	0.303 (0.129)**
Log Cum. Patent Citations ($t-3$)	-0.053 (0.054)	-0.205 (0.068)***
Log Age ($t-1$)	-0.097 (0.015)***	-0.022 (0.022)
Observations N	641,737	655,122
Pseudo R^2	0.41	0.45

This table estimates the probability of being involved in an overlapping minority stake acquisition. The dependent variable equals one if the firm was involved in such a deal in the following period. The probit model uses all firm-year observations from the pool of potential controls, but only pre-deal observation for treated firms. For treated firms with more than one overlapping minority acquisition, matching is performed on the year before the first acquisition. Variables with zero values are transformed as log(variable + 1). The lagged levels of cumulative patents and citations at $t-1$, $t-2$, and $t-3$ are included to match treated and control firms on the shape of their pre-treatment innovation trajectory; the three levels of the same cumulative stock are highly collinear, so their individual coefficients are not informative in isolation. Country, year, and industry (2-digit NAICS) fixed effects are included. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Pre-Acquisition Summary Statistics

(a) Acquirers

	Mean	10-Perc.	Median	90-Perc.	Obs.
Sales(t-1)	4.6	0.0	0.5	13	1,171
Number of Employees(t-1)	16,184	137	2,111	40,009	1,171
Age(t-1)	37	8.0	23	82	1,171
Fixed Assets(t-1)	3.7	0.0	0.3	10	1,167
Total Assets(t-1)	6.2	0.1	0.8	18	1,171
Intangible Fixed Assets(t-1)	0.8	0.0	0.0	0.9	1,151
Liquidity Ratio(t-1)	2.3	0.6	1.2	4.2	1,156
# Granted Patent Applications(t-1)	155	0.0	5.0	230	1,167
Cumulative # Granted Patent Applications(t-1)	3,174	4.0	69	3,871	1,167
# Patent Cites(t-1)	2,441	0.0	18	2,519	1,167
Cumulative # Patent Cites(t-1)	51,645	13	458	57,516	1,167

(b) Targets

	Mean	10-Perc.	Median	90-Perc.	Obs.
Sales(t-1)	2.6	0.0	0.4	6.7	2,543
Number of Employees(t-1)	10,193	105	1,568	23,613	2,543
Age(t-1)	40	7.0	30	85	2,543
Fixed Assets(t-1)	2.0	0.0	0.2	4.7	2,539
Total Assets(t-1)	3.4	0.0	0.4	8.1	2,543
Intangible Fixed Assets(t-1)	0.6	0.0	0.0	0.6	2,528
Liquidity Ratio(t-1)	2.0	0.6	1.2	3.6	2,533
# Granted Patent Applications(t-1)	76	0.0	3.0	129	2,543
Cumulative # Granted Patent Applications(t-1)	1,786	3.0	62	2,913	2,543
# Patent Cites(t-1)	1,008	0.0	10	1,272	2,543
Cumulative # Patent Cites(t-1)	27,049	10	530	33,752	2,543

This table provides summary statistics on the acquiring and target firms included in our sample. The table summarizes the values of firm-level variables for acquirers and targets in the pre-acquisition period. The Sales and the Assets are expressed in billion euro. The Liquidity Ratio is the ratio of current assets to current liabilities. The unit of observation is the value one year prior the acquisition (t-1), per firm and year. Outliers are excluded from this table by dropping the top percentile of observations in terms of Sales(t-1).

Table A4: Relative Size of Parties Pre-Acquisition

(a) Acquirers

	10-Perc.	25-Perc.	50-Perc.	75-Perc.	90-Perc.	Obs.
Ratio of Sales	0.32	1.05	4.79	18.05	69.57	742
Ratio of # of Employees	0.28	0.95	3.42	13.80	43.34	745
Ratio of Age	0.36	0.74	1.15	2.08	5.33	743
Ratio of Fixed Assets	0.36	1.14	5.12	26.05	101	741
Ratio of Total Assets	0.40	1.20	4.75	18.59	64.32	746
Ratio of Intangible Fixed Assets	0.03	0.49	3.90	33.35	209	671
Ratio of Liquidity	0.30	0.56	0.93	1.62	3.18	732
Ratio of # Granted Patents	0.00	0.01	1.04	8.00	35.10	457
Ratio of Cumulated # Granted Patents	0.08	0.50	3.38	18.50	144	743
Ratio of # Patent Cites	0.00	0.00	1.00	14.00	79.39	433
Ratio of Cumulated # Patent Cites	0.05	0.37	3.37	29.27	220	723

(b) Targets

	10-Perc.	25-Perc.	50-Perc.	75-Perc.	90-Perc.	Obs.
Ratio of Sales	0.01	0.03	0.12	0.56	1.98	915
Ratio of # of Employees	0.01	0.05	0.18	0.80	2.69	915
Ratio of Age	0.19	0.47	0.86	1.33	2.64	914
Ratio of Fixed Assets	0.00	0.02	0.12	0.63	2.15	911
Ratio of Total Assets	0.01	0.03	0.14	0.64	1.98	915
Ratio of Intangible Fixed Assets	0.00	0.01	0.09	1.06	7.70	848
Ratio of Liquidity	0.34	0.61	1.03	1.81	3.40	902
Ratio of # Granted Patents	0.00	0.00	0.04	0.50	3.24	643
Ratio of Cumulated # Granted Patents	0.00	0.03	0.28	2.01	17.22	912
Ratio of # Patent Cites	0.00	0.00	0.01	0.44	3.99	630
Ratio of Cumulated # Patent Cites	0.00	0.02	0.20	1.92	17.55	891

Panel (a) compares the targets and their respective acquirers across several variables the year before the minority stake acquisition. Similarly, Panel (b) compares the acquirers and their respective targets across several variables the year before the minority stake acquisition. This table presents the distribution (percentiles) of the ratios of values between targets and their acquirers (and between acquirers and their targets) for different variables in the pre-acquisition period (t-1). Observations for which (i) there are more than two parties in the acquisition, (ii) there are missing data for the other party in the acquisition, are excluded from this table. Outliers are excluded from this table by dropping the top percentile of observations in terms of Ratio of Sales.